DCA AS DECATEGORIFIED SHEAF QUANTIZATION

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Abstract. This paper proposes, and partially proves, a sheaf-theoretic lift of the difference-of-convex algorithm (DCA). The guiding slogan is: *DCA is the decategorification of GKS sheaf quantization* — marked [PROGRAM] throughout. By "decategorification" we mean here only the *readout* philosophy — extracting an argmin set from sheaf data — not the K_0/Euler sense standard in sheaf theory. The word is a paper-local term in this local sense throughout, including in the title; it is not used in the standard K_0 sense that sheaf theorists mean by "decategorify" (see caveat (iii) of §2). Guillermou–Kashiwara–Schapira [7] quantize Hamiltonian isotopies as equivalences of sheaf categories; we lift one DCA step to three sheaf moves: *specialization* to the tangent cone (the standard Kashiwara–Schapira construction that extracts the tangent cone of a sheaf at a point), *height convolution* $\star_t$ (convolution in the height variable only, the paper's addition rule for epigraphs), and a *readout* of the next iterate from the microsupport with $\tau > 0$ (the microsupport is the standard locus of propagation directions of a sheaf).

To a DC decomposition $f = g - h$ we attach the *ordered* pair of *epigraph sheaves* $(\mathcal{E}_g, \mathcal{E}_h)$ — the constant sheaves on the regions $\{t \geq g(x)\}$ and $\{t \geq h(x)\}$ above the graphs (Kashiwara–Schapira [5]). Three results are proved. First, the one-step comparison theorem: one DCA step equals specialization to the tangent cone, height convolution, and microsupport readout, proved in all dimensions from the dictionary and convex analysis (external-review proof). Second, the decomposition-dependence of DCA is categorified into a category of DC decompositions: gauge arrows identified up to adding an affine function, Lawvere-enriched by the Bregman cost (distances between splittings measured as worst-case movement penalties). Third, the gauge is identified with a Bregman generator (the composite-sense mirror map): a gauged DCA step is exactly a composite mirror-descent step, and re-optimizing the gauge mid-run is adaptive-mirror-map descent.

Exactly one gap is marked: functoriality of specialization along the graph of h, needed to make the iteration loop internal to sheaf data. Backward error is developed in full in [24]; a full claim/status ledger is given in §1. Nothing here claims a genuine GKS decategorification: the source and target categories, the kernel, and the comparison diagram are not constructed.

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1. Introduction

The purpose of this section is to introduce the DCA iteration, to state the sheaf dictionary in one page, and to record exactly what is proved and what is not.

The difference-of-convex algorithm (DCA), introduced by Pham Dinh and Le Thi [1] and surveyed in [2], minimizes a function written as a difference of convex functions, $f = g - h$ on $\mathbb{R}^n$ with g, h convex. One iteration is: choose a subgradient $s_k \in \partial h(x_k)$, then solve the convex subproblem

$$x_{k+1} \in \operatorname*{argmin}_x \big\{ g(x) - \langle s_k, x \rangle \big\}, \tag{DCA}$$

under a solvability hypothesis (the minimum is attained). Convergence under subanalytic/Kurdyka–Łojasiewicz hypotheses is classical [3]; boosted variants with rates appear in [4]. The algorithm famously *depends on the decomposition*: the same f with different pairs (g, h) produces different dynamics. This dependence is not a defect to be quotiented away — it is the structure §5 categorifies.

1.1. **The dictionary as consequences.** To a DC decomposition $f = g - h$ we attach the *ordered* pair of epigraph sheaves $\mathcal{D} = (\mathcal{E}_g, \mathcal{E}_h)$. Each row of the table below is one DCA move lifted to sheaves: *specialization* is the standard construction that extracts the tangent cone; *height convolution* $\star_t$ is the paper's addition rule for epigraphs (convolution in the t-variable only); the *readout* extracts the argmin set from the microsupport. The DCA datum, the linearization step, and solving the subproblem each lift to a sheaf operation, and the numerical algorithm is recovered as the readout:

DCA operation	sheaf operation
decomposition (g, h)	ordered pair $(\mathcal{E}_g, \mathcal{E}_h)$
choose $s_k \in \partial h(x_k)$	specialization $\nu_{p_k}(\mathcal{E}_h)$ + choice of supporting half-space $\mathcal{E}_{\ell_{s_k}}$
convex subproblem $\min_z(g(z) - \langle s_k, z\rangle)$	height convolution $F_{k,s} = \mathcal{E}_g \star_t \mathcal{E}_{-\ell_{s_k}}$
next iterate x_{k+1}	microsupport readout: argmin set with $\tau > 0$ (formula unpacked in §4)
change of decomposition $(G, H) \mapsto (G+C, H+C)$	simultaneous convolution $-\star_t \mathcal{E}_C$ on both components
Bregman penalty D_C	function readout of the nonlinearity sheaf $N_{C,x_k} = \mathcal{E}_C \star_t \mathcal{E}_{-c}$

The programmatic slogan is *DCA is the decategorification of GKS sheaf quantization* — marked [PROGRAM] throughout. By the paper's local convention "decategorification" means only the readout philosophy (sheaf datum $\mapsto$ argmin set), never K_0. What is proved is the one-step comparison below; the source/target categories, the kernel/functor, and the comparison diagram for a genuine GKS decategorification are not constructed here (see §4). Guillermou–Kashiwara–Schapira [7] quantize Hamiltonian isotopies to equivalences of sheaf categories; Tamarkin [6] pioneered the sheaf-theoretic approach to symplectic rigidity. To our knowledge no literature anticipates this bridge. The Legendre-transform-equals-Fourier–Sato-transform identification we use is classical Kashiwara–Schapira [8] — credited here, not claimed. D'Agnolo–Kashiwara [10] is cited only for the enhanced Fourier–Sato framework in which that identification is stated microlocally, not as the source of the identification itself.

1.2. **Main theorem and the gap.**

Theorem 1.1 (Main theorem: corrected one-step comparison — proved)**.** *Let g be finite, C^1, convex, h finite convex subanalytic on $\mathbb{R}^n$. Fix x_k and $s \in \partial h(x_k)$; put $\ell_s(z) = h(x_k) + \langle s, z - x_k\rangle$ and $F_{k,s} = \mathcal{E}_g \star_t \mathcal{E}_{-\ell_s}$. Then*

$$\pi_x\big(\{(x, t; 0, \tau) \in \mathrm{SS}(F_{k,s}) : \tau > 0\}\big) \;=\; \operatorname*{argmin}_z \{g(z) - \langle s, z\rangle\}. \tag{1}$$

If g is strictly convex and the minimum is attained, this set is a singleton. The proof is from the dictionary and convex analysis (§4); no step needs

$n = 1$. Note. *This is the same statement proved as Theorem 4.6; the duplication is intentional (statement in §1, proof in §4).*

Remark 1.2 (The gap, flagged)**.** Each sheaf step takes x_k as external input; the iteration loop is not yet an endomorphism of sheaf data. Closing it needs functoriality of $x_k \mapsto \nu_{(x_k, h(x_k))}(\mathcal{E}_h)$ along graph(h) — the single known gap, marked in §4. The internal family/iteration question is distinct from the one-step theorem above. *Feeds hub:* this theorem is the proved one-step half of the hub's [THEOREM modulo G5] [25]; the open half is exactly gap G5.

Forward references. §2 proves the dictionary; §3 computes everything on non-trivial 1D examples; §4 proves Theorem 1.1 and states the general conjectures; §5 builds the gauge category; §6 is the mirror-descent bridge; §7 lifts gauge change to convolution; §8 is backward error; §9 the tropical reading; §10 a numerical experiment; §11 the open problems with acceptance conditions.

What is proved and what is conjectured.

Claim	Status	Where
Sheaf–function dictionary	proved at support level; verified numerically on grids	§2
One-step comparison theorem, all n (external-review proof); readout corrected to $\tau > 0$ (erratum for v0.5's Def. 4.2, which returned the whole $\mathbb{R}^n$ for $\psi = x^2$); 1D baby cases (Thms. 4.8–4.13)	proved	§4
Internal family/iteration form	conjectural [PROGRAM]; one marked gap: functoriality of specialization along graph(h); source/target categories, kernel/functor, comparison diagram not constructed	§4
Gauge category of DC decompositions: arrows mod affine, Bregman cost (affine-invariant, separated, no $C \geq 0$); component inference corrected (smooth counterexample); reachable up-set, mutual reachability, weak component, groupoid completion distinct	proved as stated; separated metric on objects marked; χ monotone along $C \geq 0$ arrows (on the nose; does not descend to mod-affine classes) (same objects, arrows restricted); one monotonicity claim restricted (fixed solver and cost model)	§5
Gauge change as convolution; Bregman divergence as function readout of the nonlinearity sheaf (smooth and nonsmooth, per subgradient choice); "toward x_k" corrected to $D_C(y, x_k) \leq D_C(a, x_k)$	proved at support/function level; microlocal upgrade conjectural	§7
Gauged DCA step = composite mirror-descent step with Bregman generator $\omega = C$ (mirror map iff C Legendre); base identification Faust–Fawzi–Saunderson 2023	proved; base cited not claimed; descent for iteration-varying splittings proved under exact subproblems; novelty narrowed to quality-functional-driven choice	§6
Each floating-point step conditionally backward-exact on a gauge-perturbed pair	(a),(b) proved; (c),(f) observed; (d) proved 1D (sharp composable); (e) structural + measured 1D; unconditional exactness false (explicit counterexample); realized walk at cumulative objects marked; general n open	§8

Nothing else is claimed. Proof sketches appear only where we have them; gaps are marked, not hidden. Every statement is labelled as proved, observed, or conjectural; no proof is fabricated; original statements are kept with errata where corrected.

Remark 1.3 (Service statement: how this paper serves the hub)**.** This paper is a companion to the hub paper [25], and every result below exists to serve it. Explicitly: the dictionary (§2) and the one-step comparison theorem (Theorem 1.1) are the proved half of the hub's [THEOREM modulo G5] on DCA as sheaf quantization — the missing half is functoriality of specialization along graph(h), which is gap G5 in the hub's progress map and which this paper does not close. The gauge category (§5) and the mirror bridge (§6) develop the language the hub's DCA–sheaf subsection uses; the backward-error thread (§8) is developed in full in the companion paper [24] and appears here as the quantitative leg feeding the same hub thread. Each main result below carries a forward reference to the hub thread it feeds.

2. The corrected dictionary

The purpose of this section is to fix the sheaf vocabulary, to record the exact dictionary between DCA operations and sheaf operations, and to state the three caveats that govern everything after. As observed by Kashiwara–Schapira [5], the epigraph is the natural sheaf-theoretic encoding of a function; the dictionary below is a refinement of that observation to the DC setting, and we shall make use of it in every section that follows.

We warn the reader of two things at the outset. First, the readout is corrected here to $\tau > 0$ — v0.5's version returned the whole $\mathbb{R}^n$. Second, the K_0 reading is explicitly not what we use: K_0 here means the Grothendieck group of constructible sheaves, the standard sense in which sheaf theorists "decategorify"; our readout is a different, set-valued extraction (see caveat (iii)).

2.1. **Epigraph sheaves.** Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be finite, continuous and subanalytic (the subanalytic class is the standard tame class in which Kashiwara–Schapira specialization is defined; it contains all piecewise-polynomial and most optimization-relevant functions). Put

$$\mathcal{E}_\varphi \;=\; k_{\{(x,t):\, t \geq \varphi(x)\}} \;\in\; \mathrm{Sh}(\mathbb{R}^n_x \times \mathbb{R}_t), \tag{2}$$

the *epigraph sheaf*: the constant sheaf (over a field k) on the region above the graph of φ. The subanalytic hypothesis is not used in this section; it will be essential in §4, where we invoke Kashiwara–Schapira specialization, which needs it. For a DC pair we always work with the *ordered* pair $\mathcal{D} = (\mathcal{E}_g, \mathcal{E}_h)$; see the decomposition-dependence caveat in §2.

2.2. **Operations.** Height convolution is

$$F \star_t G \;=\; Ra_!\big(q_1^{-1}F \otimes^L q_2^{-1}G\big), \qquad a(x,s,t) = (x, s+t), \tag{3}$$

convolution in the t-variable only. The corrected dictionary is:

sheaf operation	result	function operation
$\mathcal{E}_g \star_t \mathcal{E}_h$	$\mathcal{E}_{g+h}$	sum
$\mathcal{E}_g \star_t \mathcal{E}_{-h}$	$\mathcal{E}_{g-h}$	DC difference
$\mathcal{E}_g \otimes^L \mathcal{E}_h$	$\mathcal{E}_{\max(g,h)}$	max (NOT sum)
$[\mathcal{E}_g] - [\mathcal{E}_h]$ in K_0	signed region $\mathbf{1}_{\mathrm{epi}(g)} - \mathbf{1}_{\mathrm{epi}(h)}$	remembers the pair (g, h)

The sum rule follows from the fiber description: over (x, u) the fiber is $\{s : g(x) \le s \le u - h(x)\}$, nonempty iff $u \ge g(x) + h(x)$ (cf. Courte–Shende [22], Lemma 3.13). The fibers are compact intervals, hence the addition map is proper on the support and $R^i a_! = 0$ for $i > 0$, so the identification holds as sheaves, not only at support level. The two sheaf products differ: $\star_t$ adds the heights (hence the functions), while $\otimes$ intersects the supports. The tensor is max, not sum, as shown on constants: for $g \equiv 1$, $h \equiv 2$, $\mathcal{E}_g \otimes \mathcal{E}_h = k_{[2,\infty)}$ (support = intersection), whereas $\mathrm{epi}(g+h) = [3, \infty)$.

2.3. **Caveats.**

(i) *Microsupport is conic.* The microsupport SS (standard Kashiwara–Schapira invariant) records, at each point, the codirections in which the sheaf "changes"; it is always a cone in the cotangent fibers. For smooth φ the nonzero part of $\mathrm{SS}(\mathcal{E}_\varphi)$ is $\{(x, \varphi(x); -\tau\, d\varphi(x), \tau) : \tau > 0\}$. The graph of $d\varphi$ appears only after dehomogenization (the $\tau = 1$ slice). Estimates of the form $\mathrm{SS}(F \otimes G) \subset \mathrm{SS}(F) + \mathrm{SS}(G)$ require non-characteristic hypotheses (cf. Schapira [23], Cor. 3.11).

(ii) *Euler $\neq$ Lebesgue.* The local Euler characteristic gives $\chi_{\mathrm{loc}}([\mathcal{E}_g] - [\mathcal{E}_h]) = \mathbf{1}_{t \ge g} - \mathbf{1}_{t \ge h}$. Recovering $h - g$ by $\int dt$ is a Lebesgue integral, whereas sheaf pushforward uses the Euler integral (χ_c the compactly supported version: $\chi_c([0, 1)) = 0 \neq 1$). Recovering *values* needs extra measure structure; recovering *locations* (argmin) does not. Our comparison theorem is about the dynamics (locations), hence unaffected.

(iii) *Decomposition dependence.* $1 = 1 - 0 = 2 - 1$ give *different* constructible functions $-\mathbf{1}_{[0,1)}$ vs $-\mathbf{1}_{[1,2)}$ (a constructible function is an integer linear combination of indicator functions of subanalytic sets — the standard decategorified shadow of a sheaf). Hence $[\mathcal{E}_g] - [\mathcal{E}_h]$ is not an invariant of $g - h$ alone.

The DCA datum is the ordered pair $(\mathcal{E}_g, \mathcal{E}_h)$, not the difference. Indeed DCA dynamics needs the pair: $f(x) = x^2/2$ with $(g, h) = (x^2, x^2/2)$ gives contraction ratio $1/2$, while $(3x^2/2, x^2)$ gives $2/3$. Moreover $g \mapsto [\mathcal{E}_g]$ takes function-addition to *convolution* ($\star_t$), not to K_0-addition; there is no lift of "$\mathrm{DC}(X)$ = Grothendieck group of (convex, +)" via $g \mapsto [\mathcal{E}_g]$.

3. Baby cases: everything computed on a non-trivial example

The purpose of this section is to compute the whole pipeline on one non-trivial example, to exhibit the choice space at a kink, and to supply the

toy case that every proof sketch in §4 will follow. The example to keep in mind is the following: $g(x) = x^2$ against $h(x) = x^2/4 + 1$ (smooth) and $h(x) = |x|$ (kink). At the kink, three new objects appear. The *tangent cone* is the first-order approximation of the epigraph at the point. The *choice space* is the set of candidate linearizations (slopes) the cone admits. The *microsupport fan* is the fan of codirections of the microsupport at the kink — one ray per admissible slope. We note at this stage that the 1D computations are grid/machine verified, not proved in print; the general statements they illustrate are proved in §4.

3.1. **Smooth case.** Take $g(x) = x^2$, $h(x) = x^2/4 + 1$ on $\mathbb{R}$, so $f = g - h = 3x^2/4 - 1$. Classical DCA: $h'(x) = x/2$, hence $x_{k+1} = x_k/4$. The sheaf pipeline at x_k:

(i) linearize: read the slope s_k from $\mathrm{SS}(\mathcal{E}_h)$ at $(x_k, h(x_k))$; form the tangent half-space $\mathcal{E}_{\ell_k} = k_{\{t \geq h(x_k) + s_k(x - x_k)\}}$;
(ii) convolve: $F_k = \mathcal{E}_g \star_t \mathcal{E}_{-\ell_k} = \mathcal{E}_{\psi_k}$, $\psi_k = g - \ell_k$;
(iii) read the next iterate from the microsupport: $X_{k,s} = \pi_x(\{(x, t; \xi, \tau) \in \mathrm{SS}(F_k) : \xi = 0,\ \tau > 0\})$, i.e. the zero-$x$-covector locus with the zero section excluded (= $\mathrm{Crit}(\psi_k)$ after dehomogenization).

Numerically (601×801 grid): $\mathrm{supp}(\mathcal{E}_g \star_t \mathcal{E}_h) = \mathrm{epi}(g+h)$ and $\mathrm{supp}(\mathcal{E}_g \star_t \mathcal{E}_{-h}) = \mathrm{epi}(g - h)$ hold exactly on the grid. The pipeline gives $x_{k+1} = x_k/4$ at $x_k \in \{2.0, -1.5, 0.7, 0.0\}$; twelve steps from 3.0 converge to 0, the critical point of f. Decomposition dependence is reproduced: $(x^2, x^2/2)$ gives ratio 0.5000, $(3x^2/2, x^2)$ gives 0.6667, for the same $f = x^2/2$.

3.2. **Nonsmooth case: specialization is the canonical home.** Take $g(x) = x^2$, $h(x) = |x|$. At the kink $p = (0, 0)$, the specialization (the standard Kashiwara–Schapira construction, *Sheaves on Manifolds*, Ch. IV) is

$$\nu_p(\mathcal{E}_h) \;=\; k_{C_p(\mathrm{epi}\, h)} \;=\; k_{\{w \geq |v|\}}, \tag{4}$$

the constant sheaf on the tangent cone — canonical, involving no choice. The key point is what specialization does not do: it exhibits the whole cone, not one linearization. The microsupport at the kink is the fan $\{(0, 0; \xi, \tau) : \tau > 0,\ |\xi| \leq \tau\}$. A tangent half-space $H_s = \{t \geq sx\}$ contains $\mathrm{epi}(|x|)$ iff $|s| \leq 1$ (sharp, grid-verified); hence the *choice space* is exactly

$$S_0 \;=\; \{s : \mathcal{E}_{\ell_s} \supset \mathrm{supp}\, \nu_p(\mathcal{E}_h)\} \;=\; [-1, 1] \;=\; \partial h(0), \tag{5}$$

and each subgradient corresponds to a ray of the microsupport fan.

The per-choice pipeline recovers the DCA update exactly on the test slopes $\{-1, -0.5, 0, 0.5, 1\}$: for each such s, the zero-locus of $\mathrm{SS}(\mathcal{E}_{x^2 - sx})$ is $x = s/2$ to machine precision. The "bad" choice $s = 0$ stalls at the kink — a critical non-minimizer — and the sheaf pipeline gives the same result. At smooth points (e.g. $x_k = 1$) the specialization is already the unique half-space, so smooth and nonsmooth are unified: smooth = singleton choice space, nonsmooth = cone plus supporting hyperplanes.

Remark 3.1 (Dead ends recorded)**.** Nearby cycles is *not* the home for the nonsmooth lift: it needs a chosen smoothing family $h_\varepsilon \to h$, and the limit depends on the family — non-canonical. Stalks are the wrong object: stalks of $\nu_p(\mathcal{E}_h)$ are 0/1 membership and select no linearization.

Remark 3.2 (Honest ledger for §3)**.** The 1D computations are grid/machine verified, not proved in print; the general-n statements they illustrate are proved (Theorem A/B baby cases) or conjectured (§4) with the readout corrected to $\tau > 0$ throughout.

4. The comparison theorems

The purpose of this section is to define the sheaf DCA step, to prove the one-step comparison in all dimensions, and to isolate the single known gap. We begin by elaborating on the notions sketched in the introduction: specialization, already canonical in Kashiwara–Schapira [5], becomes here the family of linearizations, and the microsupport readout of §2 becomes the next iterate. The key point is that no step of the proof needs $n = 1$: the toy case of §3 was one-dimensional only for computability.

4.1. **Definitions.** Let $g, h : \mathbb{R}^n \to \mathbb{R}$ be finite continuous convex (possibly nonsmooth), $\mathcal{D} = (\mathcal{E}_g, \mathcal{E}_h)$ the DC sheaf datum.

Definition 4.1 (Sheaf linearization)**.** Fix x_k and $p_k = (x_k, h(x_k))$. The *specialization* $\nu_{p_k}(\mathcal{E}_h)$ is the Kashiwara–Schapira specialization of $\mathcal{E}_h = k_{\mathrm{epi}(h)}$ at p_k; it is the canonical family of linearizations. Throughout, ν_p denotes the one-sided specialization ν_p^+ (the standard Kashiwara–Schapira convention). The identification $\nu_{p_k}(\mathcal{E}_h) \simeq k_{C_{p_k}(\mathrm{epi}\, h)}$ is not definitional: it is the specialization-to-tangent-cone lemma (Lemma 4.9 below), proved by a direct normal-deformation computation for the convex class; the general Kashiwara–Schapira theorem [5] needs the subanalytic hypotheses stated there. A *linearization* is a supporting half-space $\mathcal{E}_{\ell_s} = k_{\{t \geq h(x_k) + \langle s, x - x_k\rangle\}}$ for s in the *choice space* S_k^{choice} of supporting slopes (= rays of the microsupport fan of $\nu_{p_k}(\mathcal{E}_h)$ at the vertex).

Definition 4.2 (Sheaf DCA step, set-valued)**.** Given $s \in S_k^{\mathrm{choice}}$: (i) form $\mathcal{E}_{\ell_s}$; (ii) form $F_{k,s} = \mathcal{E}_g \star_t \mathcal{E}_{-\ell_s} = \mathcal{E}_{\psi_{k,s}}$, $\psi_{k,s} = g - \ell_s$; (iii) put $X_{k,s} = \pi_x(\{(x, t; \xi, \tau) \in \mathrm{SS}(F_{k,s}) : \xi = 0,\ \tau > 0\})$, the zero-$x$-covector locus with the zero section excluded. The *readout* is $\mathrm{Read}(\mathcal{D}, x_k, s) = X_{k,s} \subset \mathbb{R}^n$ *as a set.*

Remark 4.3 (Instantiation)**.** In the toy case of §3, S_k^{choice} is the singleton $\{x_k/2\}$ for the smooth $h(x) = x^2/4 + 1$, and the interval $[-1, 1] = \partial h(0)$ for the kink $h(x) = |x|$; the readout is $\{x_k/4\}$ and $\{s/2\}$, respectively. The abstract definitions above say exactly this, with the example's numbers replaced by the general mechanism.

Remark 4.4 (Erratum: the $\tau > 0$ correction)**.** v0.5's readout omitted $\tau > 0$. Zero covectors belong to $\mathrm{SS}(\mathcal{E}_\psi)$ over the whole support (every x has $t \geq \psi(x)$), so the uncorrected readout returns all of $\mathbb{R}^n$: for $\psi = x^2$ it gives $\mathbb{R}$ instead of $\{0\}$. The hub paper's Definition 12.18 already had $\tau \neq 0$; the two papers are now synchronized. All statements below use the corrected readout.

Remark 4.5 (Set-valued correction)**.** An earlier formulation spoke of "the next point". A critic exhibited $g(x) = \max(0, x)^2$, $h(x) = \max(0, |x| - 1)$: at $s = 0$ the argmin is the *interval* $(-\infty, 0]$, and the sheaf zero-locus equals it as a set. Hence the pipeline is stated set-valued; point-level DCA is the pipeline *plus* a selection rule, exactly as in classical DCA. The unbounded case is faithful too: empty zero-locus $\iff$ the subproblem is unsolvable.

Theorem 4.6 (One-step comparison, all n — proved)**.** *Let g be finite, C^1, convex, h finite convex subanalytic on $\mathbb{R}^n$. Fix x_k and $s \in \partial h(x_k)$; put $\ell_s(z) = h(x_k) + \langle s, z - x_k\rangle$ and $F_{k,s} = \mathcal{E}_g \star_t \mathcal{E}_{-\ell_s}$. Then*

$$\pi_x\big(\{(x, t; 0, \tau) \in \mathrm{SS}(F_{k,s}) : \tau > 0\}\big) \;=\; \operatorname*{argmin}_z \{g(z) - \langle s, z\rangle\}. \tag{6}$$

If g is strictly convex and the minimum is attained, this set is a singleton.

Proof supplied by external review. [1] Convolution gives $F_{k,s} = \mathcal{E}_{g-\ell_s}$. The microsupport formula for a smooth epigraph gives the covector $(-\tau(\nabla g - s), \tau)$ at the graph. With $\tau > 0$ imposed, vanishing of the x-covector is equivalent to $\nabla g = s$, equivalent to a global minimizer by convexity. For nonsmooth h, the supporting slopes of the tangent cone are exactly $\partial h(x_k)$. No step needs $n = 1$. □

Remark 4.7. This is the proved form of the identical statement in Theorem 1.1. It is a consequence of the dictionary and convex analysis, not a new theorem about GKS quantization. The proof above was supplied by external review and is reproduced verbatim. It closes the one-step forms (Theorem 4.11 for the smooth case at agent level, and the baby cases); the internal family/iteration forms (Conjecture 4.12 and Conjecture 4.14) remain open (§5 and the gap below).

4.2. **Theorem A: smooth case.**

Theorem 4.8 (Theorem A, baby case — proved)**.** *Let $g(x) = x^2$, $h(x) = x^2/4 + 1$ on $\mathbb{R}$, and assume each subproblem $\min_x(g - \ell_k)$ is attained. Then for the unique $s_k = h'(x_k)$,*

$$\mathrm{Read}(\mathcal{D}, x_k, s_k) \;=\; \big\{\operatorname*{argmin}_x \left(g(x) - \langle h'(x_k), x\rangle\right)\big\} \;=\; \{x_k/4\}, \tag{7}$$

i.e. the sheaf pipeline recovers the classical DCA update exactly, step for step.

[1] i.e., proved by an independent reviewer during drafting; the argument below is reproduced verbatim.

Proof sketch. This proof follows the outline of the toy case in §3.1, and checks two ingredients. The first is that smooth specialization picks no choice: h smooth at x_k implies $\nu_{p_k}(\mathcal{E}_h)$ is already the tangent half-space, so $S_k^{\text{choice}} = \{h'(x_k)\}$ is a singleton (standard fact about specialization). At a technical level, one needs the fiber computation of §2 to get $\mathcal{E}_g \star_t \mathcal{E}_{-\ell_k} = \mathcal{E}_{g-\ell_k}$, and for strictly convex C^1 ψ the zero-x-covector locus of $\mathrm{SS}(\mathcal{E}_\psi)$ is $\{\psi' = 0\}$, the unique minimizer. Putting this all together proves the exact step-for-step recovery. Numerically verified at four base points and over twelve iterates. □

Lemma 4.9 (Specialization-to-tangent-cone, convex class)**.** *Let $h : \mathbb{R}^n \to \mathbb{R}$ be finite convex (hence continuous), $Z = \mathrm{epi}(h) = \{(x,t) : t \geq h(x)\} \subset \mathbb{R}^n \times \mathbb{R}$, $p = (x_0, h(x_0)) \in \partial Z$, and $\mathcal{E}_h = k_Z$.*

(a) *Z is closed and convex; subanalytic if h is.*
(b) *The tangent cone (the contingent and Clarke cones coincide for convex Z) is $C_p(Z) = \mathrm{epi}(h'(x_0;\cdot)) = \{(v,\tau) : \tau \geq h'(x_0;v)\}$, where $h'(x_0;v) = \lim_{t\downarrow 0} \frac{h(x_0+tv)-h(x_0)}{t}$ exists, is finite, positively homogeneous and convex (hence sublinear). If h is C^1 at x_0 then $C_p(Z) = H_{x_0} := \{(v,\tau) : \tau \geq \langle \nabla h(x_0), v\rangle\}$.*
(c) *With the one-sided specialization ν_p^+ (the standard convention of Kashiwara–Schapira [5]), $\nu_p(\mathcal{E}_h) \simeq k_{C_p(Z)}$.*

[Proved — agent level; pending author review; Lean verification pending.] (A-049.)

Proof. (a) Convexity: for $(x_i,t_i) \in Z$, $\lambda \in [0,1]$, $\lambda t_1 + (1-\lambda)t_2 \geq \lambda h(x_1) + (1-\lambda)h(x_2) \geq h(\lambda x_1 + (1-\lambda)x_2)$. Closedness: $Z = (t - h(x))^{-1}([0,\infty))$ with h continuous. If h is subanalytic, so is $\mathrm{graph}(h)$, and $Z = \Phi(\mathrm{graph}(h) \times [0,\infty))$ with the linear map $\Phi(x,s,u) = (x, s+u)$ proper on this set; hence Z is subanalytic as a proper direct image of a closed subanalytic set. (b) For $t > 0$, $\phi(t) := \frac{h(x_0+tv)-h(x_0)}{t}$ is nondecreasing in t (slope monotonicity of convex functions) and bounded below (h is locally Lipschitz), so $h'(x_0;v) = \lim_{t\downarrow 0} \phi(t)$ exists and is finite; it is positively homogeneous and a pointwise limit of convex functions, hence convex. For ($\subset$): if $(v,\tau) \in C_p(Z)$ via $t_k \downarrow 0$, $(v_k,\tau_k) \to (v,\tau)$ with $h(x_0) + t_k\tau_k \geq h(x_0 + t_k v_k)$, dividing by t_k and letting $k \to \infty$ gives $\tau \geq h'(x_0;v)$ (the difference term vanishes by local Lipschitz). For ($\supset$): if $\tau \geq h'(x_0;v)$, with $\alpha_k = \frac{h(x_0+t_kv)-h(x_0)}{t_k} \to h'(x_0;v)$ and $\tau_k = \max(\alpha_k, \tau) \to \tau$, the points $(x_0 + t_k v, h(x_0) + t_k\tau_k) \in Z$ witness $(v,\tau) \in C_p(Z)$. If h is C^1 at x_0 then $h'(x_0;v) = \langle \nabla h(x_0), v\rangle$. (c) Translate $p = 0$ (so $0 \in Z$). With the one-sided normal deformation $\widetilde{X}_0 = \{(y,t) : t \geq 0\}$, $p(y,t) = ty$, $s(v) = (v,0)$, $j^+ : \{t > 0\} \hookrightarrow \widetilde{X}_0$, $\nu_0^+(F) = s^{-1}Rj_*^+(p^{-1}F|_{\{t>0\}})$; then $p^{-1}(k_Z)|_{\{t>0\}} = k_{\widetilde{Z}^+}$ with $\widetilde{Z}^+ = \bigcup_{t>0}(Z_t \times \{t\})$, $Z_t = \frac{1}{t}Z$. Monotonicity (using only convexity of Z and $0 \in Z$): $0 < t_1 < t_2$ implies $Z_{t_1} \supset Z_{t_2}$. Put $C = \overline{\bigcup_{t>0} Z_t}$ (the tangent cone of the convex set Z at 0). Computing sections over a basis of open balls B, $\Gamma(B; \nu_0^+ k_Z) =$

$\varinjlim_{\epsilon\downarrow 0} H^0(\mathcal{W}_{B,\epsilon}; k)$ with $\mathcal{W}_{B,\epsilon} = \{(y,t) : y \in B, 0 < t < \epsilon, y \in Z_t\}$: if $B \cap C = \varnothing$ then $\mathcal{W}_{B,\epsilon} = \varnothing$; if $B \cap C \neq \varnothing$ then $\mathcal{W}_{B,\epsilon} \neq \varnothing$ and is path-connected (monotonicity lets one descend in t and move across the convex slices), so $\Gamma(B; \nu_0^+ k_Z) = k = \Gamma(B; k_C)$, naturally in B. Hence $\nu_0^+(k_Z) \simeq k_C$ as sheaves. □

Remark 4.10. Part (c) uses only the definition of the normal deformation plus convexity (the monotonicity of the scalings Z_t) — no Kashiwara–Schapira theorem is invoked, and no subanalyticity is needed for the convex class. Subanalyticity is required only for the general specialization theorem of [5]. The one-sided convention ($\Omega = \{t > 0\}$, the standard KS one) is essential: the two-sided version does not give k_C (e.g. $Z = [0, \infty) \subset \mathbb{R}$ has stalk k at $v = -1 \notin C$ coming from the $t < 0$ side).

Theorem 4.11 (Theorem A*, one-step — proved at agent level)**.** *Let g, h be C^1, strictly convex, finite and* subanalytic *on $\mathbb{R}^n$, with each subproblem $\min_z\{g(z) - \langle \nabla h(x_k), z\rangle\}$ attained. Then* $\mathrm{Read}(\mathcal{D}, x_k, \nabla h(x_k))$ *is the singleton $\{\mathrm{argmin}_x(g(x) - \langle \nabla h(x_k), x\rangle)\}$, i.e. the sheaf pipeline recovers the classical DCA update exactly, step for step. [Proved — agent level; pending author review; Lean verification pending.] The specialization step is Lemma 4.9: $\nu_{p_k}(\mathcal{E}_h) \simeq k_{H_{x_k}}$, proved by a direct normal-deformation computation for the convex class (no subanalyticity needed there). Subanalyticity is retained as an explicit hypothesis for the general Kashiwara–Schapira machinery invoked elsewhere in the paper.*

Proof sketch. This is the n-dimensional form of the proof sketch of Theorem 4.8, with the specialization-to-tangent-cone lemma proved directly for the convex class. First, h is C^1 at x_k, so $\nu_{p_k}(\mathcal{E}_h)$ is already the tangent half-space: for the convex set $Z = \mathrm{epi}(h)$ one has $\nu_p(k_Z) \simeq k_{C_p(Z)}$ via the one-sided normal deformation and the scaling monotonicity of convex sets, and the tangent cone is $C_p(Z) = \{(v,\tau) : \tau \geq \langle \nabla h(x_k), v\rangle\}$. Hence smooth specialization picks no choice and $S_k^{\mathrm{choice}} = \{\nabla h(x_k)\}$ is a singleton. Second, at the fiber level, $\mathcal{E}_g \star_t \mathcal{E}_{-\ell_k} = \mathcal{E}_{g-\ell_k}$ by the fiber computation of §2, and for strictly convex C^1 ψ the zero-x-covector locus of $\mathrm{SS}(\mathcal{E}_\psi)$ is $\{\psi' = 0\}$, the unique minimizer. Putting these together, the sheaf readout is the singleton $\{\mathrm{argmin}_x(g(x) - \langle \nabla h(x_k), x\rangle)\}$, i.e. the exact classical DCA step at x_k. □

Conjecture 4.12 (Sheaf iteration coincides with classical DCA — conjectural)**.** *An internal sheaf iteration of the one-step pipeline coincides with classical DCA as sequences. [CONJECTURAL.] See §11(1) for the functoriality gap this requires.*

4.3. **Theorem B: nonsmooth case.**

Theorem 4.13 (Theorem B, baby case — proved)**.** *Let $g(x) = x^2$, $h(x) = |x|$ on $\mathbb{R}$, subproblems attained. Then:*

(i) *$\nu_{p_k}(\mathcal{E}_h)$ is the canonical family of linearizations; it does not and must not pick one.*

(ii) $S_k^{\text{choice}} = \partial h(x_k)$ *exactly (supporting hyperplanes of the tangent cone = rays of the microsupport fan).*
(iii) *For* each $s \in S_k^{\text{choice}}$, $\text{Read}(\mathcal{D}, x_k, s) = \{\text{argmin}_x(g(x) - \langle s, x\rangle)\}$ *as sets — the per-choice pipeline recovers DCA exactly, including bad choices.*
(iv) *At smooth points this reduces to Theorem 4.8.*

Proof sketch. The new points here are the canonical specialization and the fan computation. The first is that specialization is canonical: (i) is the standard Kashiwara–Schapira specialization for closed subanalytic sets, via the normal deformation — it exhibits the family and must not pick one linearization. At a technical level, one computes the fan: (ii) at the kink, $\text{SS}(\mathcal{E}_h)_{(0,0)} = \{(0,0;\xi,\tau) : \tau > 0, |\xi| \le \tau\}$ and $H_s \supset \text{epi}(|x|) \iff |s| \le 1$, both grid-verified sharp; (iii) per choice s, apply Theorem 4.6 to the affine ℓ_s, verified for five choices to machine precision; (iv) at $x_k = 1$: $\nu = k_{\{w \ge v\}}$, unique $s = 1$. □

Conjecture 4.14 (Theorem B, general — conjectural)**.** *Let g be C^1 strictly convex, h convex (possibly nonsmooth) on $\mathbb{R}^n$, subproblems attained. Then (i)–(iv) above hold with $S_k^{\text{choice}} = \partial h(x_k)$, the readout being set-valued.*

4.4. **The gap.**

Remark 4.15 (Gap: functoriality — sharpened)**.** Each sheaf step takes the point x_k as external input; the iteration loop is not yet an endomorphism of sheaf data. A fully internal iteration needs the specialization *along* $\text{graph}(h)$ as a family, i.e. functoriality of $x_k \mapsto \nu_{(x_k, h(x_k))}(\mathcal{E}_h)$. This is the single known gap in the comparison theorem, now sharpened: the "i.e." above was loose language. Specialization along $\text{graph}(h)$ in the literal Kashiwara–Schapira sense forgets the slope: its fiber is $k_{[0,\infty)}$, independent of x as an abstract sheaf. The family the gap actually needs is the universal at-point family $\{\nu_{p_x}\}$, $p_x = (x, h(x))$ — and nonzero inter-fiber morphisms do exist: for slopes $s_x = \nabla h(x)$ and $s_{x'} = \nabla h(x')$ differing ($H_x = \{(v,\tau) : \tau \ge \langle s_x, v\rangle\}$ the supporting half-space), the canonical maps $k_{H_x} \twoheadrightarrow k_{H_x \cap H_{x'}} \hookrightarrow k_{H_{x'}}$ compose to a nonzero morphism (at $p = (0, \tau_0)$, $\tau_0 > 0$, both maps are isomorphisms on stalks $k \to k$). [Proved — agent level; pending author review; Lean verification pending.] What is missing is not morphisms but a sheaf-functorial *choice* coherent with the iteration: a canonical (choice-free) iteration endofunctor Φ_K exists and iterates the DCA sequence exactly at exact level; it is not purely sheaf-functorial because the Argmin readout is an external set-theoretic extraction, not a sheaf functor. See §11(1).

Remark 4.16 (The slogan is a program, not a theorem)**.** "DCA is the decategorification of GKS quantization" is marked [PROGRAM]. GKS quantize Hamiltonian isotopies by kernels and equivalences; our pipeline has iterate-dependent specialization, one subgradient choice, and an Argmin readout.

A genuine decategorification claim would need: source and target categories, the kernel/functor, natural transformations, and an explicit comparison diagram. None are constructed here. The microsupport readout is furthermore not a K_0-decategorification in the sense defined in §2.3(iii).

4.5. **Boundaries inherited from DCA.** The lift is faithful, so it inherits DCA's hypotheses as precise sheaf statements (numbering follows the hub paper's boundary list [25]): (B2) non-attainment $\iff$ empty zero-locus; (B3) flat bottoms $\iff$ interval-valued zero-locus, needing a selection rule; (B5) argmin *locations* need no measure structure (Euler vs. Lebesgue only affects values); (B6) the loop-external gap above.

Remark 4.17 (Honest ledger for §4)**.** Theorem 4.6 is proved (external-review proof); the 1D baby cases are proved; the internal iteration and the GKS decategorification slogan are open programs. Nothing in this section proves a new theorem about GKS quantization.

5. The category of DC decompositions

The purpose of this section is to turn decomposition-dependence into a category, to put a metric on its arrows, and to correct the component inference. The decomposition-dependence recorded in §2 (caveat (iii)) is not a bug but a structure: many splittings of the same F, with arrows given by adding a common convex function.

We categorify it, following the gauge analysis of the authors' monograph [12] (Prop. 5.5.1, Theorem A.1: the gauge change $(G, H) \mapsto (G + C, H + C)$ alters the DCA majorant by exactly the Bregman divergence D_C). We warn the reader that "gauge" here means representational redundancy (same F, many (G, H)), not a physical gauge theory. A *thin* category (groupoid, preorder) is one with at most one arrow between any two objects — so thinness says the arrow, when it exists, is forced. A *Lawvere metric space* (Lawvere [9]) is a metric space with symmetry and separation dropped: distances may be infinite and $d(x, y) = 0$ need not imply $x = y$.

Theorem 5.1 (The DC groupoid — proved)**.** *Fix a domain Ω and $F : \Omega \to \mathbb{R}$. Objects: ordered pairs (G, H) of convex functions with $F = G - H$. A morphism $D : (G_1, H_1) \to (G_2, H_2)$ is the unique function with $G_2 - G_1 = H_2 - H_1 = D$; composition is pointwise addition, identity is 0. This is a thin groupoid: all axioms are the axioms of function addition, inverses are $-D$, Hom-sets are at most singletons. In words: changing the gauge by D is reversible and unique, when D is allowed to be any function.*

Theorem 5.2 (The gauge preorder — proved)**.** *Same objects as Theorem 5.1. A morphism is a* gauge morphism *if $D = C$ is convex and differentiable, taken* modulo affine functions *(the thesis's original formulation). Composition is pointwise addition; inverses are generally absent ($-C$ is concave). This is a thin category, i.e. a preorder: arrows only* add *convexity.*

Lemma 5.3 (Affine invariance of Bregman divergence — proved)**.** *For affine $\ell(z) = \langle a, z\rangle + b$, $D_{C+\ell} \equiv D_C$: the extra gradient term cancels in $(C+\ell)(z) - (C+\ell)(x) - \langle \nabla C(x) + a, z - x\rangle$. Hence the Bregman divergence descends to the mod-affine quotient — the single fact the earlier bit-cost Ξ lacked.*

Proof. $D_{C+\ell}(z,x) - D_C(z,x) = \ell(z) - \ell(x) - \langle a, z - x\rangle \equiv 0$, since ℓ affine gives $\ell(z) - \ell(x) = \langle a, z - x\rangle$. The divergence is therefore well-defined on the mod-affine quotient. □

Remark 5.4 (History of the cost — why two costs are kept)**.** v0.1 took arrows mod affine with the bit-cost $\Xi(C) = 2 \sup C/|F|$, which is not affine-invariant: $[0] = [3x + 1]$ gives 0.0 vs 7.0 on $\Omega = [2, 3]$, $F = x^2 - |x|$, and the inf-over-representatives repair diverges to $-\infty$. v0.2 repaired this by taking arrows on the nose, the affine-freedom surviving as a congruence through which the majorant-field functor factors (since $D_{C+\ell} = D_C$). Decision (canonicity). Theorem 5.6 keeps the thesis's original arrows and changes the cost instead, and the division of labor is now explicit. The Bregman cost d is the *canonical* enrichment: affine-invariant, separated, and defined on the full smooth gauge category over bounded Ω with $F \not\equiv 0$ — no $C \geq 0$ restriction, and algorithmically meaningful (the worst-case movement penalty of one gauge step).

The bit-cost Ξ is retained as the thesis's original counting measure wherever d is undefined or vacuous: the $F \equiv 0$ fiber (the motivating component-disconnectedness witness, where d divides by zero) and unbounded Ω (where $\sup D_C = \infty$ for every non-affine C, collapsing d to the $\{0, \infty\}$ dichotomy). Hence the backward-error gauge-fit sequence of Theorem 8.1(e), which lives on $\mathbb{R}^n$, is measured in Ξ, not d; the Ξ-bound there is formal ($\infty \leq \infty$ for the sequence's quadratic gauges) and becomes quantitative on any bounded Ω containing the trajectory. The two enrichments are not interchangeable; each is used at its own site.

Remark 5.5 (Correction: incomparability $\not\Rightarrow$ different components)**.** v0.5 inferred different gauge components from the absence of a direct gauge arrow. That inference is wrong. On $\Omega = [-2, 2]$ with $F \equiv 1$,

$$P_1 = (x^4 + 1, x^4), \qquad P_2 = (x^2 + 1, x^2) : \tag{8}$$

$x^2 - x^4$ and $x^4 - x^2$ are both non-convex on Ω, so no gauge arrow joins P_1, P_2 in either direction. But

$$P_1 \xrightarrow{+x^2} P_* = (x^4 + x^2 + 1,\ x^4 + x^2) \xleftarrow{+x^4} P_2, \tag{9}$$

so P_1, P_2 lie in the *same* weak component, by fully smooth gauges. In general two smooth splittings (G_i, H_i) of the same F have the common upper bound $(G_1 + G_2, H_1 + G_2)$ with gauges G_2, G_1. Four distinct notions must not be conflated: the *reachable up-set* $\uparrow P$ (splittings reachable by gauge arrows from P), *mutual reachability*, the *weak component* (zigzags allowed), and the *groupoid completion*.

For $F \equiv 0$, (x^2, x^2) and $(|x|, |x|)$ admit no direct arrow either way, but are weakly connected through $(x^2+|x|, x^2+|x|)$ when nonsmooth convex gauges are allowed; with smooth-only gauges the zigzag is obstructed and the singularity invariant (the $|x|$ kink) separates them. The scope (PL/smooth/full-convex) must be stated with every component claim.

Theorem 5.6 (Bregman cost is the enrichment — proved)**.** *For a gauge arrow $[C]$ (mod affine, Theorem 5.2), put*

$$d([C]) \;=\; \frac{\sup_{z,x\in\Omega} D_C(z,x)}{\sup_\Omega |F|} \tag{10}$$

(the Bregman cost; standing hypotheses: Ω bounded, F not $\equiv 0$ on Ω (so $\sup_\Omega |F| > 0$), C differentiable *on Ω — the Bregman divergence $D_C(z,x) = C(z) - C(x) - \langle \nabla C(x), z - x\rangle$ needs ∇C — and D_C extends continuously to $\bar\Omega \times \bar\Omega$). Then:*

(i) *d is well-defined on the quotient — no inf-over-representatives repair, no $-\infty$ pathology;*
(ii) *$d([0]) = 0$, and $d([C]) = 0 \iff [C]$ is the identity arrow — the metric is* separated*;*
(iii) *$d([C_1] + [C_2]) \le d([C_1]) + d([C_2])$.*

Hence the gauge preorder, with the thesis's original mod-affine arrows, is enriched over $([0,\infty], +, 0)$ — a Lawvere metric space — with Hom $= \infty$ *where no gauge arrow exists (across weak components). Unlike the earlier bit-cost Ξ, this needs no $C \ge 0$ restriction: $D_C \ge 0$ for every convex* differentiable *C (the first-order convexity inequality $C(z) \ge C(x) + \langle \nabla C(x), z - x\rangle$; the nonsmooth case is per-choice, Remark 5.10 below), so the enrichment lives on the full* smooth *gauge category. In words: the distance between two splittings is the worst-case movement penalty one gauge step adds to DCA's subproblem — equivalently, the diameter of Ω in C's own dualistic geometry (Amari–Nagaoka [17]).*

Proof. (i) is Lemma 5.3 ($D_{C+\ell} = D_C$ exactly). (ii): $D_C \equiv 0 \iff C$ is affine — one x_0 suffices: $C(z) = C(x_0) + \langle \nabla C(x_0), z - x_0\rangle$ for all z. (iii): pointwise $D_{C_1+C_2} = D_{C_1} + D_{C_2}$, and sup of a sum $\le$ sum of sups. Numerical verification on $\Omega = [2,3]$, $F = x^2 - |x|$: $\max |D_{C+\ell} - D_C| = 3.77\times10^{-15}$ (affine invariance); the Ξ-hole witness $x^2/2$ vs $x^2/2+3x+1$ gives equal cost $0.0833 = 0.0833$; pointwise additivity to 10^{-15}; triangle $0.2500 \le 0.0833 + 0.1667$ tight; cost(affine) $\approx 3 \times 10^{-16}$, cost$(x^2/2) = 0.0833 > 0$ (separation). □

Definition 5.7 (Affine quotient on objects)**.** $\sim_{\text{aff}}$ identifies (G, H) with $(G + \ell, H + \ell)$ for affine ℓ; it is a genuine equivalence relation compatible with the arrows.

Remark 5.8 (Object quotient — resolved)**.** The separation in Theorem 5.6 is on *arrows* mod affine. Quotienting objects by common affine shift (Definition above), the zero-two-sided-distance pathology disappears by definition

[Proved — agent level; pending author review; Lean verification pending.] What remains is a full two-sided separated metric on the quotient objects: two-way reachability in the smooth thin category forces the gauges affine (a convex C with $C + C'$ affine and C, C' convex is affine), hence trivial after the quotient, so a nontrivial object metric would need zigzag infima on the groupoid completion, for which the "sums of near-affine stay near-affine" lemma is missing [Open].

Remark 5.9 (Lawvere length — marked)**.** The cumulative Bregman work $\sum_k D_{C_k}(x_{k+1}, x_k)$ is a trajectory-dependent action, not the length of the fixed arrow cost $\sup D_C / \sup |F|$; calling it Lawvere length needs a state-enriched category, not yet constructed. We keep the cumulative work with this marking.

Remark 5.10 (Nonsmooth arrows need per-selection costs)**.** Theorem 5.6 needs C differentiable: for merely subdifferentiable C the cost of a gauge arrow is not well-defined — e.g. on $\Omega = [0, 1]$, $F = x^2 - |x|$, $d_\sigma([|x|]) = 8.0/4.0/0.0$ for the selections with $\sigma(0) = -1/0/1$ (and $\sigma(x) = 1$ for $x > 0$), verified numerically with $D^\sigma_{|x|} \geq 0$ throughout. The honest fan-style extension (cf. Theorem 7.3, which is *local*: one x_k, one $s \in \partial C(x_k)$) is one cost $d_\sigma([C]) = \sup_{z,x} D^\sigma_C(z, x) / \sup |F|$ per *selection* $\sigma(x) \in \partial C(x)$ — a single frozen s does not give $D^s_C \geq 0$ in general, since nonnegativity needs $s \in \partial C(x)$ at each x. Separation survives per selection; subadditivity across selections $d_\sigma([C_1] + [C_2]) \leq d_{\sigma_1}([C_1]) + d_{\sigma_2}([C_2])$ is **open**: it needs the sum-rule decomposition $\sigma = \sigma_1 + \sigma_2$ of selections, which is not worked out. We keep d for smooth gauges and use d_σ at kinks; a selection-free single cost for nonsmooth arrows is left open.

Theorem 5.11 (Quality functionals are functors — proved on $C \geq 0$; one claim restricted)**.** *The subtraction condition number, in absolute-value form*

$$\chi_{DC}(x) \;=\; \frac{|G(x)| + |H(x)|}{|F(x)|}, \qquad F(x) \neq 0 \tag{11}$$

(thesis Def. 5.5.2; the form $(G + H)/|F| \geq 1$ is valid only for nonnegative components, and χ is undefined on the $F \equiv 0$ fiber), satisfies $\chi_C(x) = \chi_0(x) + 2C(x)/|F(x)|$ along gauge arrows in the nonnegative case.

Repair note (red team). The inequality $\chi_C(x) \geq \chi_0(x)$ needs $C(x) \geq 0$ pointwise and is false in general: for $F = x^2 - |x|$ on $\Omega = [2, 3]$ and $C(x) = (x - 2.5)^2 - 1$ (convex, differentiable, $C(2.5) = -1$), $\chi(2.5)$ goes $2.333 \to 1.800$ along the gauge arrow. Hence χ is monotone along arrows with $C \geq 0$ on Ω (on the nose, for actual representatives) — a one-sided monotonicity that does not descend to mod-affine arrow classes, not a functor on the whole preorder. Hessian monotonicity $\nabla^2(G + C) \succeq \nabla^2 G$ needs only $\nabla^2 C \succeq 0$ and holds for all arrows. □

The four-term quality functional $\mathcal{Q}(G, H)$ (thesis Def. 5.8.2) is expected to be likewise monotone along $C \geq 0$ arrows, conditional on the thesis's control of each term (not verified here). Citation status. Thesis Def. 5.8.2 is a "proposed diagnostic" (thesis Remark 5.8.3), not a standard definition; cite it with that status, never as established. Its cancellation rank can serve as an alternative tiebreak for the trace quality Q_{tr}, agreeing with the trace ordering on the quadratic ray. Its "cost of convex subproblem" term is restricted, not verified in general*: adding convexity can* lower *the condition number — e.g. Hessian* $\mathrm{diag}(1, 100)$ *plus gauge Hessian* $\mathrm{diag}(99, 0)$ *gives* $100I$*, condition number* $100 \to 1$ *— and Loewner monotonicity does not imply solver iteration count increases. The claim needs a fixed solver and cost model; marked [restricted — solver model required].*

Theorem 5.12 (Majorant field: functor; DCA: relational at iterates — proved at model level)**.** *Let H be differentiable and $M_{G,H}(z|x) = G(z) - H(x) - \langle \nabla H(x), z - x\rangle$ be the DCA majorant field. The assignment $(G, H) \mapsto M_{G,H}$ is a strict functor to the preorder of majorant fields under pointwise $\leq$, sending a gauge arrow C to $M_{G,H} \leq M_{G,H} + D_C$ — this is thesis Prop. 5.5.1 (Theorem A.1). The iterate stage $M \mapsto \mathrm{argmin}_z M(z|x)$ is relational (set-valued) by the correction of §4, so* $\mathrm{DCA} = \mathrm{Argmin} \circ \mathrm{MajField}$ *is strict at the model level and relational at the iterate level. The majorant field also descends to the object quotient $\sim_{\mathrm{aff}}$ and separates its classes: for affine $\ell(z) = \langle a, z\rangle + b$, $M_{G+\ell,H+\ell} \equiv M_{G,H}$ pointwise (the affine parts cancel between the function value and the linear term), and the map $[(G, H)] \mapsto M_{G,H}$ is injective — the diagonal recovers $F = F'$ via $M(x|x) = G(x) - H(x)$, and if $M_{G_1,H_1} \equiv M_{G_2,H_2}$ then for each fixed x, $G_1 - G_2 = a_x^1 - a_x^2$ is affine in z (with $a_x^i(z) = H_i(x) + \langle \nabla H_i(x), z - x\rangle$), hence $G_1 - G_2 = \ell$ for a fixed affine ℓ, and $F_1 = F_2$ forces $H_1 - H_2 = \ell$ as well — so the majorant field is a complete invariant of the $\sim_{\mathrm{aff}}$-class [Proved — agent level; pending author review; Lean verification pending.].*[2]

Remark 5.13 (Why optimal representation is hard)**.** "Best splitting" = χ-minimal object: minimal elements of the preorder, not initial ones. The thesis's centred optimal pair and the naive propagated pair are not joined

[2]This theorem is cited by §8(d) for the model/iterate boundary; it is kept because it records the exact functoriality boundary the paper respects: strict at the model level, relational (set-valued) at the iterate level, consistent with the set-valued correction of §4. The boundary is intrinsic, not a place awaiting more effort: upward, the model-to-iterate lift is not well-defined — two valid float32 evaluations of the same gradient give bitwise-different iterates (149/150 steps on the 1D run); downward, iterate behavior factors through a strictly coarser quotient — model perturbations below the rounding quantum are bit-invisible at iterate level [Proved — agent level (numerical); pending author review; Lean verification pending.]. Quantitatively, on the 2D double-well run, (G, H) versus $(G + \varepsilon\|z\|^2, H + \varepsilon\|z\|^2)$ are bit-identical for $\varepsilon = 10^{-12}$ and $\varepsilon = 10^{-9}$ (0/80 steps differ), begin to flip at $\varepsilon = 10^{-7}$ (69/80 differ), and differ everywhere at $\varepsilon = 10^{-5}$: the visibility threshold sits exactly at the rounding quantum $\varepsilon \sim u = 2^{-24}$ against $O(1\text{–}10)$ quantities [Proved — agent level (numerical); pending author review; Lean verification pending.].

by a gauge arrow in the smooth-gauge category, so no within-category arrow reaches the optimum — this is why the optimal representation problem (thesis Remark 5.8.7) is hard. The gauge category explains the difficulty and bounds the within-up-set optimum (thesis Thm. 5.6.4, quoted in full as Theorem A.2: the within-up-set Pareto frontier is a straight line). [Component scope: "different components" here means no direct gauge arrow in the stated gauge class; weak-component membership is the finer question corrected above.]

Remark 5.14 (Honest ledger for §5)**.** The groupoid, preorder, affine invariance, and Bregman enrichment are proved as stated. Component claims carry an explicit gauge-class scope; the separated-metric-on-objects and state-enriched-category steps are marked open. Solver-cost monotonicity is narrowed as follows: for gradient descent with step-size rule $\eta = 1/L$ (where L is the Lipschitz constant of the current objective's gradient) and the certified iteration bound $\bar{T} = \kappa \log(\mathrm{diam}(\Omega)/\varepsilon)$ (with $\kappa = L/\mu$ the condition number, $\mathrm{diam}(\Omega)$ the domain diameter, ε the target accuracy), adding a convex gauge increment Ψ along a gauge arrow at fixed outer iterate x_k does not increase the certified cost if and only if $\kappa_\Psi \le \kappa_\Phi$ [Proved — agent level; pending author review; Lean verification pending.], where Φ is the current subproblem objective and $\kappa_\Phi = L_\Phi/\mu_\Phi$ its condition number. In particular a scalar ray in a well-conditioned generator (e.g. $C_0 = \frac{1}{2}\|\cdot\|^2$, $\kappa_{C_0} = 1$) is nonincreasing. Even with the solver and cost model fixed, a convex-only increment ($\mu_\Psi = 0$) strictly increases the cost: measured $\kappa : 100 \to 50100$, iterations $2062 \to 1038226$ [Refuted in the un-narrowed form]. "Fixed solver" must include the step-size rule — a literally fixed η can diverge on the gauged problem — and monotonicity is along gauge arrows at fixed x_k, not along the iterate sequence. *Feeds hub:* this category is the language in which the hub's DCA–sheaf subsection speaks of decomposition-dependence; the Bregman cost is the quantitative form of the hub's "movement costs".

6. Gauge is a Bregman generator: the mirror-descent bridge

The purpose of this section is to identify the gauge with the generator of a composite mirror-descent step, to prove descent for iteration-varying splittings, and to narrow the novelty claim to what is actually new. Our approach is closest to Faust–Fawzi–Saunderson [13], who prove DCA ≡ Bregman proximal point for fixed decompositions, and can be compared with the adaptive proximal methods of Duchi–Hazan–Singer [20]; all these points of view are variants of Beck–Teboulle mirror descent [14]. What those papers do *not* have is a free gauge: we treat the Bregman penalty's generator as a *choice*, not as fixed data.

The gauge C is not only a categorical arrow (§5) and a sheaf convolution (§7): it is a *Bregman generator* — the convex function whose divergence penalizes the mirror-descent subproblem. We state the bridge precisely, with an honest ledger of what is prior art.

Proposition 6.1 (Gauge = Bregman generator — proved as composite identity)**.** *The DCA subproblem with a differentiable gauge C is a composite mirror-descent step with Bregman generator $\omega = C$. From Prop. 5.5.1 (Theorem A.1),*

(12)
$$M_{G+C,H+C}(z|x_k) \;=\; G(z)-H(x_k)-\langle s_k, z-x_k\rangle+D_C(z,x_k), \qquad s_k \in \partial H(x_k),$$

and dropping z-independent terms the subproblem is

$$x_{k+1} \in \operatorname*{argmin}_z \big\{G(z) - \langle s_k, z\rangle + D_C(z, x_k)\big\}. \tag{13}$$

This matches composite mirror descent $x_{k+1} = \operatorname{argmin}_z\{\langle g_k, z\rangle + r(z) + \eta^{-1}D_\omega(z, x_k)\}$ with $g_k = -s_k$, $r = G$, $\omega = C$, $\eta = 1$ exactly.

Remark 6.2 ("Mirror map" in the classical sense)**.** The identity above is exact as a composite Bregman subproblem. To call C a *mirror map* in the usual sense needs Legendre/strict-convexity hypotheses giving nondegenerate mirror geometry; an affine or zero gauge does not. We therefore use "Bregman generator" for the composite identity throughout, and reserve "mirror map" for the classical Legendre meaning.

Proposition 6.3 (Descent for varying splittings — proved)**.** *If each round chooses any convex splitting $f = G_k - H_k$, $s_k \in \partial H_k(x_k)$, and solves the subproblem exactly, then*

$$f(x_{k+1}) \;\le\; G_k(x_{k+1}) - H_k(x_k) - \langle s_k, x_{k+1} - x_k\rangle \;\le\; f(x_k). \tag{14}$$

Hence descent is not an open problem under these hypotheses, even though the splitting changes between rounds.

Proof supplied by external review. x_{k+1} minimizes $G_k(\cdot)-\langle s_k, \cdot\rangle$, so $G_k(x_{k+1})-\langle s_k, x_{k+1}\rangle \le G_k(x_k) - \langle s_k, x_k\rangle$; subtracting $H_k(x_k)$ and using $f(x_k) = G_k(x_k) - H_k(x_k)$ and convexity of H_k gives the chain. □

Remark 6.4. What remains genuinely open: convergence/rates, uniform coercivity/curvature, inexact solves, and the benefit of quality-driven adaptation. The proof above was supplied by external review and is reproduced verbatim.

Prior art, in detail (cited, not claimed). The equivalence of DCA with the Bregman proximal-point algorithm (with respect to the second function) is proved in Faust–Fawzi–Saunderson [13], §2.3; their main focus is convergence analysis. Their Remark 1 adds that when the first function is also differentiable, DCA is equivalently mirror descent with the Bregman divergence generated by it. The mirror-descent mechanism itself is Beck–Teboulle [14].

Remark 6.5 (Framework identification: DCA = BBT Bregman proximal gradient)**.** The rate of the fixed-gauge theorem is the Bauschke–Bolte–Teboulle [15] Bregman proximal gradient rate for $\min\{-H+G\}$ with $(f, g, h, L) = (-H, G, C, 1)$. The L-smad condition ($Lh-f$ convex) is $C+H$ convex, which

holds automatically — no extra hypothesis is needed. Scope notes: convex relative smoothness does not apply directly ($F = G - H$ is nonconvex); triangle scaling/acceleration does not apply (DCA has no momentum). [Proved — agent level; pending author review; Lean verification pending.]

Remark 6.6 (The gauge generalizes the stepsize)**.** $\eta^{-1} D_\omega = D_{\eta^{-1}\omega}$, and $\eta^{-1}\omega$ is convex iff ω is: choosing the stepsize η is the scalar case of choosing the gauge. The thesis's slogan becomes: choosing a decomposition = choosing a proximal regularizer = choosing a Bregman generator.

Remark 6.7 (Adaptive-mirror-map descent — narrowed novelty)**.** Re-optimizing the gauge every m iterations against a decomposition-quality functional (e.g. χ_{DC} or $\mathcal{Q}(G, H)$), with jumps to incomparable splittings, is mirror descent with an *iteration-varying Bregman generator* $\omega_k = C_k$; numerically this improves the conditioning (see §8's discussion). Sharpness warning: the gauge must not change too fast. With $C_k(z) = \frac{2^k}{2} z^2$ ($L_k = 2^k \to \infty$ while $\mu_k = 2^k \geq 1$ stays bounded away from 0; L_k the gradient Lipschitz constant, μ_k the strong-convexity modulus of C_k) DCA converges to a non-stationary point ($x_k \to x_0/4.768\ldots \neq 0$ for $G(z) = z^2/2$, $H = 0$); the threshold is sharp, $x_k \to 0 \iff \sum_k 1/L_k = \infty$. So $\sup_k L_k < \infty$ is a genuine hypothesis for stationarity, not a technicality, while $\inf_k \mu_k > 0$ is what keeps the $O(1/K)$ rate. [Proved — agent level; pending author review; Lean verification pending.]

Precedents for adapting the proximal geometry mid-run exist and are cited: Duchi–Hazan–Singer adapt the proximal function during the run, diagonal and full-matrix [20], and McMahan analyzes adaptive regularizers and mirror descent [21]. The only literature precedent for changing the *decomposition* mid-run is Wu–Sun [18], whose changing set of convex functions is *scheduled* by the ODE discretization, not *optimized* against a quality functional — the closest existing precedent, cited not claimed. What appears new is the combination: re-optimizing the map itself mid-run against a decomposition-quality functional, with the categorical framing (Bregman generator = gauge arrow; §5's enrichment cost = the worst-case mirror-descent movement penalty — the thesis title, "Movement Costs", made categorical). Tan et al. learn a mirror map once, offline, and fix it during the run [16].

Remark 6.8 (Correction: what gauge jumps can and cannot do)**.** A pure gauge jump $(G + C, H + C)$ can never lower *pointwise* χ: for $C \geq 0$ on Ω, $\chi_C(x) = \chi_0(x) + 2C(x)/|F(x)| \geq \chi_0(x)$, and a 1-D λ-search over $(G + \lambda C, H + \lambda C)$ always returns $\lambda^* = 0$.

Stronger: χ is not even a function on the gauge class, let alone on an object quotient [Refuted]. On $F = x^2 - |x|$ over $\Omega = [2, 3]$, the gauge jump by $C(x) = (x - 2.5)^2 - 1$ sends $\chi(2.5)$ from 2.333 to 1.800, and the affine shift $\ell(x) = 3x + 1$ sends it to 6.867. The functoriality of χ that survives is

one-sided only: monotonicity along $C \geq 0$ arrows (stated above), with no descent.

The χ drops seen in the gauge-jumping numerics therefore came from *incomparability* jumps (to the centered pair, which admits no direct gauge arrow from the naive pair), never from gauge moves. Gauge moves are for DCA *conditioning*: the Bregman term D_C enters the majorant via Prop. 5.5.1 (Theorem A.1), acting as a proximal penalty on the subproblem — not for χ reduction.

Adaptive-gauge DCA thus has two distinct mechanisms, which the quality functional should score separately: incomparability jumps choose a better splitting (lower χ); within-up-set gauge moves condition the subproblem. Scope of the adaptation refutation: numerically, proximal damping — adaptive local-curvature gauges $C_k = \tau\,\mathrm{diag}(\nabla^2 G)$ — does *not* accelerate DCA (on the 2D double well, $\tau = 2$ costs $3.5\times$ the outer steps and $3.1\times$ the inner Newton work of plain DCA). The gauge acts as proximal damping, not as a preconditioner: stronger damping means smaller steps. This refutes only proximal-damping speedup; the incomparability-jump mechanism (splitting choice lowering χ) is separate and remains open. [Refuted (proximal sense) — agent level; pending author review; Lean verification pending.]

Terminology. Following the literature survey in the campaign notes: "adaptive DCA" in the literature means adaptive line search or inner termination, never decomposition adaptation — our algorithm is called *adaptive-gauge DCA* explicitly; likewise "adaptive-mirror-map descent" is used in full, never shortened to "adaptive mirror descent".

Remark 6.9 (Honest ledger for §6)**.** Proposition 6.1 is exact as a composite identity; "mirror map" in the classical sense needs Legendre hypotheses. Descent for varying splittings is proved (Proposition 6.3); fixed-gauge rates ($O(1/K)$), the adaptive-gauge characterization ($O(1/K)$ under $\inf_k \mu_k > 0$; stationarity needs $\sup_k L_k < \infty$), and the inexactness threshold ($\sum \varepsilon_k^2 < \infty$ preserves the rate) are proved at agent level; the proximal-damping adaptation benefit is refuted (numerical); the linear rate for DC stays open. Novelty is claimed only for quality-functional-driven splitting choice, with adaptive-proximal precedents cited.

7. Gauge change is convolution; Bregman is its readout

The purpose of this section is to lift gauge change to sheaves, to identify the Bregman divergence as the readout of the *nonlinearity sheaf* (the paper's term for the gauge minus its own linearization — the nonlinear part of C at x_k), and to correct the proximal "toward x_k" reading. The basis is Prop. 5.5.1 of the thesis (Theorem A.1; gauge change alters the majorant by D_C); what is new here is the sheaf lift, which makes the Bregman term a convolution. The key point is that the sheaf side never needed differentiability — only the classical Bregman formula did.

Theorem 7.1 (Gauge lifts to simultaneous convolution — proved)**.** *A gauge morphism $C : (G, H) \mapsto (G + C, H + C)$ lifts to sheaves as the endofunctor $- \star_t \mathcal{E}_C$ applied to* both *components:*

$$(\mathcal{E}_g, \mathcal{E}_h) \longmapsto (\mathcal{E}_g \star_t \mathcal{E}_C,\ \mathcal{E}_h \star_t \mathcal{E}_C) \ = \ (\mathcal{E}_{g+C},\ \mathcal{E}_{h+C}). \tag{15}$$

Here C is assumed differentiable (so $\nabla C(x_k)$ exists); the nonsmooth case, per subgradient choice, is Theorem 7.3. Linearization shifts accordingly: with $c(z) = C(x_k) + \langle \nabla C(x_k), z - x_k \rangle$, $\mathcal{E}_{\ell'} = \mathcal{E}_\ell \star_t \mathcal{E}_c$ and slopes shift by $s' = s + \nabla C(x_k)$ (i.e. $\partial(h + C) = \partial h + \nabla C$ read off the microsupport fan).

Proof sketch. The equality $\mathcal{E}_g \star_t \mathcal{E}_C = \mathcal{E}_{g+C}$ is the sum rule of the §2 dictionary: $\star_t$ adds heights, and the fiber over (x, u) is $\{s : g(x) \le s \le u - C(x)\}$, nonempty iff $u \ge g(x) + C(x)$ (cf. Courte–Shende [22], Lemma 3.13, as recorded in §2). Applying the same rule to the affine function c, whose slope is $\nabla C(x_k)$, gives $\mathcal{E}_\ell \star_t \mathcal{E}_c = \mathcal{E}_{\ell'}$ and the slope shift; differentiability of C is used only here. □

Theorem 7.2 (Commutativity; Bregman = readout of the nonlinearity sheaf — proved)**.** *Put $N_{C,x_k} = \mathcal{E}_C \star_t \mathcal{E}_{-c}$ (the "nonlinearity sheaf": the gauge minus its own linearization at x_k). Then gauge-then-pipeline equals pipeline-then-convolve:*

$$F' \ = \ F \star_t N_{C,x_k}, \qquad i.e. \quad \psi' = \psi + D_C(\,\cdot\,, x_k) \tag{16}$$

exactly *(verified to 3.6×10^{-15} on $g = x^2$, $h = |x|$, $C = x^2/2$, including the kink, per subgradient choice).*

Proof sketch. $F' = \mathcal{E}_{g'} \star_t \mathcal{E}_{-\ell'} = (\mathcal{E}_g \star_t \mathcal{E}_C) \star_t (\mathcal{E}_{-\ell} \star_t \mathcal{E}_{-c}) = (\mathcal{E}_g \star_t \mathcal{E}_{-\ell}) \star_t (\mathcal{E}_C \star_t \mathcal{E}_{-c}) = F \star_t N$, by associativity/commutativity of $\star_t$. The function readout of N_{C,x_k} is $C - c = D_C(\cdot, x_k)$. □

Theorem 7.3 (Nonsmooth gauges, per subgradient choice — proved)**.** *Let C be convex and merely subdifferentiable, x_k fixed, $s \in \partial C(x_k)$. Put $c_s(z) = C(x_k) + \langle s, z - x_k \rangle$, $N^s_{C,x_k} = \mathcal{E}_C \star_t \mathcal{E}_{-c_s}$ and $D^s_C(z, x_k) = C(z) - c_s(z) \ge 0$. Then per choice s, gauge-then-pipeline equals pipeline-then-convolve with N^s_{C,x_k}, and the function readout is exactly the generalized Bregman divergence:*

$$M'(z|x_k)\,[s] \ = \ M(z|x_k) + D^s_C(z, x_k). \tag{17}$$

The sheaf side never needed differentiability ($\mathcal{E}_C = k_{\mathrm{epi}(C)}$ is defined for any continuous C); only the classical Bregman formula did.

Proof sketch. Per choice s, c_s is affine, so the proof of Theorem 7.2 applies verbatim with c replaced by c_s: $F'_s = F \star_t N^s_{C,x_k}$, whose function shadow is $C - c_s = D^s_C(\cdot, x_k) \ge 0$. Verified to 1.8×10^{-15} for $C(x) = |x|$ at the kink $x_k = 0$, $s \in \{-1, 0, 0.5, 1\}$, via the microsupport-fan mechanism of §3.2 (each s is a fan ray). □

Theorem 7.4 (The proximal mechanism via the nonlinearity sheaf — corrected)**.** *The identity of Theorem 7.2 is exact; its* interpretation *is corrected here. If a minimizes ψ and y minimizes $\psi + D_C(\cdot, x_k)$, then*

$$D_C(y, x_k) \;\le\; D_C(a, x_k), \tag{18}$$

i.e. the gauged argmin is "toward x_k" in the Bregman *geometry of C — not in Euclidean distance. Indeed, with an affine gauge $D_C \equiv 0$ on the whole domain, so the zero-covector locus of the nonlinearity sheaf is not just x_k; penalty and Euclidean distance must be distinguished.*

Even a strictly convex gauge need not move the Euclidean argmin closer at $n > 1$: with $x_k = 0$, $a = (0, 1)$, $\psi(z) = \frac{1}{2}(z - a)^\top H(z - a)$, $H = \operatorname{diag}(1, 100)$, $C(z) = \frac{1}{2}z^\top Qz$, $Q = \left(\begin{smallmatrix} 2 & 9 \\ 9 & 82 \end{smallmatrix}\right) > 0$, the old argmin is a with $\|a\| = 1$ while the new argmin is $y = (H + Q)^{-1}Ha = (-60/31,\, 20/31)$ with $\|y\|^2 = 4000/961 > 1$.

The thesis's "choosing a decomposition = choosing a proximal regulariser" becomes: choosing a decomposition = choosing which N to convolve the majorant sheaf with. The same holds per subgradient choice for nonsmooth C (Theorem 7.3), with $D_C^s \ge 0$.

Proof sketch of the corrected inequality. y minimizes $\psi + D_C(\cdot, x_k)$, so $\psi(y) + D_C(y, x_k) \le \psi(a) + D_C(a, x_k)$; since a minimizes ψ, $\psi(y) \ge \psi(a)$, and the D_C inequality follows. □

Conjecture 7.5 (Microlocal upgrade — conjectural)**.** *Theorems 7.1–7.3 lift to microsupport estimates under the non-characteristic hypotheses of §2.3(i), expected for subanalytic convex data.*

Remark 7.6 (Honest ledger for §7)**.** Gauge-as-convolution and Bregman-as-readout are proved at the support/function level (smooth and per-choice nonsmooth); the microlocal upgrade is conjectural. The proximal "toward x_k" reading is corrected to the Bregman geometry $D_C(y, x_k) \le D_C(a, x_k)$ (Theorem 7.4); Euclidean closeness is false at $n > 1$.

8. Backward error: the gauge-fit sequence of a DCA run

The purpose of this section is to pose Wilkinson's question for the DC algorithm, to define the gauge-perturbed pair that answers it, and to isolate the feasibility condition on which the answer depends. Throughout, u denotes the unit roundoff. We begin by elaborating on the backward-error thread sketched in the introduction.

Wilkinson's question: for what *perturbed* problem is the computed iterate exact? For DCA the answer is a gauge perturbation. Each floating-point step is *conditionally* backward-exact on the gauge-perturbed pair $(G + C_e, H + C_e)$ — i.e. a tiny gauge arrow in §5's category, exact exactly when the gauge fit exists (the feasibility condition of Theorem 8.1(d); infeasible steps keep only the tilt part) — so a whole run yields a *gauge-fit sequence* $\{C_k\}$

whose cumulative Bregman work measures the total rounding cost. (Terminology throughout this section follows the companion paper [24], which develops the backward-error analysis in full; the *formal cumulative gauge* is $\Sigma_K = \sum_{k<K} C_k$, and a *realized walk* — a run whose steps are DCA steps of the cumulative pairs — is closed in the negative (see the honest ledger below). Yardstick convention for "realized": the word is used with three yardsticks — exact structural $\delta_{\text{struct}} = 0$ (the characterization), at the rounding-noise scale ($\delta_{\text{real}} \lesssim$ quantum), and the counted-step $\delta_{\text{struct}} \leq 10^{-9}$; each claim below states which yardstick it uses.) Throughout this section, dM_k denotes the *model-work scale* of step k — the energy scale of the exact DCA subproblem at x_k, the signal against which the fitted gauge's rounding noise is measured. This explains DCA's robustness despite doubly-exponential conditioning: the *steps* are conditionally backward-stable; only the *evaluation* of $F = G - H$ carries the forward error χu.

Theorem 8.1 (Backward-error DCA theorem)**.** *Let (G, H) be DC, H differentiable at x, and let $\tilde{s} = \mathrm{fl}(\nabla H(x)) = s + e$ be the computed subgradient and $\tilde{x}_+$ the computed approximate minimizer of $\tilde{M}(z) = G(z) - \langle \tilde{s}, z\rangle$. Put $d = \tilde{x}_+ - x$.*

(a) ***[PROVED]*** *For $\sigma \geq 0$, $C_{e,\sigma}(z) = \langle e, z\rangle + \frac{\sigma}{2}\|z - x\|^2$ is convex, $\nabla(H + C_{e,\sigma})(x) = \tilde{s}$, and $M_{G+C_{e,\sigma},H+C_{e,\sigma}}(z|x) = M_{G,H}(z|x) + \frac{\sigma}{2}\|z - x\|^2$ (Prop. 5.5.1, Theorem A.1; the linear part cancels in D_C).*

(b) ***[PROVED, conditional]*** *If $x_+(\sigma^*) := \operatorname{argmin}_z M(z) + \frac{\sigma^*}{2}\|z - x\|^2$ equals $\tilde{x}_+$, then $\tilde{x}_+$ is* backward-exact *from x for the gauge-perturbed pair $(G + C_{e,\sigma^*}, H + C_{e,\sigma^*})$.*

(c) ***[OBSERVED]*** *Such a σ^* exists with σ-*match error *$\|x_+(\sigma^*) - \tilde{x}_+\| \lesssim$ solver tolerance (the distance between the fitted gauge's exact step and the computed step): to rounding (often $\sigma^* = 0$) for Newton on smooth problems; approximately for derivative-free solvers while progressing; nonexistent when $u\|G\| \gg dM$ (see (f)).*

(d) ***[PROVED, 1D; $n > 1$ PROVED — agent level; pending author review; Lean verification pending]*** *The composition bound, on the double well. Per step, with r_k the model residual and $d_k = \tilde{x}_+ - x$ the step, the* conditional backward-error certificate *— the best feasible gauge fit — has $D_{C_k}(\tilde{x}_+, x) = \max(r_k d_k, 0)/2 \leq (u\varsigma_k|d_k|)/2$, where $|r_k| \leq u\varsigma_k$ with $\varsigma_k = 1.77|s_r| + 1.32|x_{k+1}|G''(x_{k+1})$ (tilt + Newton-stagnation, where s_r is the residual slope scale of the companion paper; the Newton part is a textbook bound [11]). This ς_k is exactly the roundoff scale S_k of the companion paper [24], quoted in full: $S_k = 1.77|s_r| + 1.32|x_{k+1}|G''(x_{k+1})$. The composable quantity is D_{C_k}, not σ_k^* ($\sigma^* = r_k/d_k$ is ill-behaved on $d_k \to 0$ steps while $D_k = r_k d_k/2 \to 0$ gracefully, and vanishes at fixed points). This is not an ad-hoc convention: writing $e_k = \tilde{s}_k - s_k$ for the subgradient evaluation error, $d_k = y_k - x_k$ for the step (here $y_k = \tilde{x}_+$ is the computed new iterate), and $r_k = s_k - \nabla G(y_k)$ for the model residual, the*

gauge $C_k(z) = \langle e_k, z\rangle + \frac{1}{2}(z-x_k)^T Q_k (z-x_k)$ *with* $Q_k = r_k r_k^T/(d_k^T r_k)$ *when* $d_k^T r_k > 0$ *and* $Q_k = 0$ *otherwise is convex* ($Q_k \succeq 0$) *and satisfies* $D_{C_k}(y_k, x_k) = \frac{1}{2}\max(d_k^T r_k, 0)$ *in every case — the genuine Bregman divergence of a convex gauge, tilt-only when the fit is infeasible [Proved — agent level; pending author review; Lean verification pending.] The gauge-fit sequence* $\{C_k\}$ *of (e) therefore has content even on infeasible steps. The choice of* D_{C_k} *over* σ^* *is forced, not convenient: no uniform bound for the fitted curvature exists as a function of* $\|r_k\|$ *alone — with fixed* $\bar{r} > 0$ *and* $d_k = 2^{-k} \to 0^+$, *every step is feasible yet* $\|Q_k\|_{\mathrm{op}} = \bar{r}2^k \to \infty$ *(in 1D,* $|\sigma_k^*| = \bar{r}2^k \to \infty$*); a uniform bound would need the feasibility scalar* $\tau_k := d_k^T r_k$ *bounded away from zero, which cannot hold along a converging run [Refuted]. The sharp substitute is the trade-off identity* $\|Q_k\|_{\mathrm{op}} \cdot D_{C_k} = \|r_k\|^2/2 \le u^2\varsigma_k^2/2$ *(exact algebra, verified to* 5.05×10^{-29} *in float64): the curvature blows up exactly where the Bregman work vanishes, their product staying at rounding scale. This closes the "large curvature, tiny work" observation by theorem — the per-step bound* $\|Q_k\|_{\mathrm{op}} \le u^2\varsigma_k^2/\tau_k$ *exists but carries the per-step* τ_k *[Proved — agent level; pending author review; Lean verification pending.]. Summing,* $R_K := \sum_{k<K} D_{C_k} \le (u/2)(\max_{k<K}\varsigma_k)\,\mathrm{TV}_K$, $\mathrm{TV}_K = \sum_k |d_k|$*: measured* 2.79×10^{-8} *against a bound* 8.83×10^{-7} *(30× slack) on the 1D double well* $F = x^4 - 2x^2$ *with Newton subproblems in float32 (150 steps). Scope: 1D, smooth, Newton; the feasibility condition* $r_k d_k > 0$ *(which makes the certificate backward-exact) is the rare case (16% of progressing steps) and the* $\sigma^* = 0$ *fallback the norm — in 1D the fit content is the size, now proved. Method caveat on refit: the fit-or-zero rule* $\tilde{\sigma} = \max(\tilde{r}/\tilde{d}, 0)$ *(*$\tilde{r}$ *the computed residual,* $\tilde{d}$ *the computed step) applied to the computed residual has positive feedback through damping — at near-stall steps* $|\tilde{d}|$ *is of order one float32 ulp while* $|\tilde{r}|$ *grows with the mass, so* $\tilde{\sigma} = \tilde{r}/\tilde{d}$ *explodes; iterating the refit to self-consistency diverges rather than converges. Any "iterate refit to self-consistency" proposal must face this mechanism. A 2D separable-double-well run (80 steps, float32 Newton) shows the near-threshold behavior the companion paper's feasibility certificate predicts: feasible steps with* $d_k^T r_k \downarrow 0^+$ *give large fitted curvature (*$\|Q_k\|_{\mathrm{op}}$ *up to* 3.7*) with tiny Bregman work (*$D_{C_k} = \frac{1}{2}d_k^T r_k \sim 10^{-14}$*) — large curvature, tiny work — so the fit-or-zero gauge is correct but fragile at the feasibility boundary* $d_k^T r_k > 0$ *[Proved — agent level (numerical); pending author review; Lean verification pending.]. In* $n > 1$ *the size bound lifts intact via Theorem 4.1(d) of the companion paper* [24]*, which we quote in full since theorem numbers shift across versions:*

Theorem 4.1(d) (Composability) of [24]. *Put* $D_{C_k} := \frac{1}{2}\max(d_k^T r_k, 0)$. *Then*

$$R_K := \sum_{k<K} D_{C_k} \le \frac{u}{2}\Big(\max_{k<K} S_k\Big)\,\mathrm{TV}_K, \qquad \mathrm{TV}_K := \sum_{k<K}\|d_k\|,$$

with no alignment hypothesis and no feasibility hypothesis. The lift is one line: $D_{C_k} = \frac{1}{2}\max(d_k^T r_k, 0) \le \frac{1}{2}\|d_k\|\,\|r_k\| \le \frac{u}{2}\varsigma_k\|d_k\|$ *(Cauchy–Schwarz plus the n-dimensional roundoff estimate* $\|r_k\| \lesssim u\varsigma_k$ *[Proved — agent level; pending author review; Lean verification pending]). With* $D_{C_k} := \frac{1}{2}\max(d_k^T r_k, 0)$,

$$R_K := \sum_{k<K} D_{C_k} \le \frac{u}{2}\Big(\max_{k<K} \varsigma_k\Big)\,\mathrm{TV}_K, \qquad \mathrm{TV}_K := \sum_{k<K}\|d_k\|,$$

with no alignment hypothesis and no feasibility hypothesis. The n-dimensional roundoff estimate $\|r_k\| \lesssim u\varsigma_k$ *on which this rests is now [Proved — agent level; pending author review; Lean verification pending] in the companion paper (Theorem 4.1, "On ρ and c", Theorem A.4):* $\|r_k\| \le uS_k$ *with* $\varsigma_k = c_1\|s_k\| + c_2\|y_k\|\,\|\nabla^2 G(y_k)\|_{\mathrm{op}}$, *via the residual decomposition* $r_k = -e - g$ *(tilt error plus solver residual). The certified ratio* $\max_k\|r_k\|/(uS_k) \le 1$ *holds numerically at* $n = 3, 5, 10$. *This estimate is the quantitative form of the model/iterate boundary of Theorem 5.12: the computed majorant* $\tilde{M}(z) = G(z) - \langle \tilde{s}_k, z\rangle$ *differs from the model majorant* $M(z|x_k) = G(z) - \langle s_k, z\rangle$ *by a tilt* $-\langle e_k, z\rangle$ *(where* $e_k = \tilde{s}_k - s_k$ *is the subgradient evaluation error and u the unit roundoff), with* $\|e_k\| \le c_1 u\|s_k\|$ *(measured* $\max\|e_k\|/(u\|s_k\|) = 0.84$*), and the computed minimizer* y_k *differs from the exact model minimizer* x_k^+ *by* $\|x_k^+ - y_k\| \le \|r_k\|/\mu \le uS_k/\mu$ *(where* μ *is the strong-convexity constant of M; measured* $\max = 5.96 \times 10^{-8}$*, one float32 quantum, ratio to the bound* ≤ 0.36*, zero violations in 150 steps) [Proved — agent level (numerical); pending author review; Lean verification pending.]. In n dimensions the bound upgrades to trajectory-local form: with* $\bar{\mu}_k = \lambda_{\min}\Big(\int_0^1 \nabla^2 G(y_k + t(x_k^+ - y_k))\,dt\Big)$ *the strong-convexity constant of the mean Hessian along the segment* $[y_k, x_k^+]$ *(no global convexity used),* $\|x_k^+ - y_k\| \le \|r_k\|/\bar{\mu}_k \le uS_k/\bar{\mu}_k$*, verified with zero violations on a 2D run and sharp to* 10^{-9} *(ratio* 0.9999999974 *attained at* $k = 24$*; the trajectory-local* μ *is up to* $2.9\times$ *tighter than the global one on the run) [Proved — agent level (numerical); pending author review; Lean verification pending.]. The estimate holds under an explicit assumption set: G is* C^2 *with* $\nabla^2 G \succ 0$ *on the trajectory (smooth steps only), the Newton solver stops by one of four named rules (exact float32 fixed point, relative stagnation, exactly-zero computed gradient, or gradient-stall), and the iteration cap* $\mathrm{maxit} = 500$ *is excluded — it is a fifth exit of the code, at which the returned*

y is arbitrary, so the estimate is conditional on non-maxit termination. The cap never fired on the validated trajectories: 0/1500 probe steps, and at most 11 *Newton iterations in the tested configs [Observation]. For gradient-stall we assume* stall-faithfulness*: the computed gradient at a stall is dominated by evaluation noise. The gradient-stall path has now fired and been verified numerically. In a 48-run probe (float32 Newton on G of quartic-plus-quadratic form with condition numbers* 10–10000*), gradient-stall fired in* 5/48 *runs, only at condition number* ≥ 1000*; at each stall the computed gradient was at most* 0.065 *of the estimated evaluation-noise floor* $\eta = \gamma_{m_g}(\|\nabla G(y)\| + \|\tilde{s}\|)$ *— over* 15× *below the noise — so stall-faithfulness holds in the strong form, not merely up to an* $O(1)$ *factor [Proved — agent level (numerical); pending author review; Lean verification pending.]. The constants* c_1, c_2 *are measured per (problem, code) and certified by the ratio test. Tight* universal *analytic values do not exist [Refuted]: the valid analytic constant counts roundings per component, so two correct implementations of the same* ∇G *give different bounds, and no universal constant can be tight for both. For a fixed code, an explicit constant in* γ_m*-form (the standard unit-roundoff factor for m per-component roundings) exists [Proved — agent level; pending author review; Lean verification pending.] but it is intrinsically loose — worst-case rounding alignment, not analysis — at* 12–28× *for* c_1 *(at* $n = 3, 5, 10$*) and* 1.5–9× *for* c_2 *(certified ratio* ≤ 0.045 *at* $n = 3, 5, 10$*). At kinks* $\nabla^2 G$ *does not exist and the estimate does not apply. The alignment* $\cos\theta$ *governs only the gauge curvature (companion Theorem 4.1(b), Theorem A.3, conditional on* $d^T r > 0$*), not the fit coefficient (companion Theorem 4.1(a), ibid.) nor the size bound above.*

(e) ***[PROVED structurally; measured quantitatively]*** *The* gauge-fit sequence $\{C_k\}$*. The K per-step fitted gauges determine the* formal cumulative gauge $\Sigma_K = \sum_{k<K} C_k$*, i.e. the arrow* $(G, H) \to (G + \Sigma_K, H + \Sigma_K)$ *in the gauge category. Every* C_k *is convex by construction (*$\sigma_k^* \geq 0$ *via* $\max(\cdot, 0)$*), so* Σ_K *is convex: the gauge-fit sequence cannot leave its weak component, by construction. The sequence's cumulative Bregman work is* $R_K = \sum_k D_{C_k}$ *of (d) — affine-invariant and composable, unlike the* Ξ*-length, which is vacuous (*$\infty \leq \infty$*) on* $\mathbb{R}^n$ *for the sequence's quadratic gauges and tilt-dominated even on bounded* Ω *(measured* Ξ*-length* 4.74×10^{-2} *vs* D_C*-length* 2.79×10^{-8} *on the trajectory hull). Quantitative, on the 1D double-well run of (d): the sequence stays uniformly* $\geq$ 1127× *below the* gauge-dominance boundary $w = D_{C_k}/dM_k = 1$ *(measured* $\max_k w_k = 8.87 \times 10^{-4}$*; gauge curvature* $< 0.2\%$ *of problem curvature), and its gauges move* χ *by* $< 2 \times 10^{-8}$ *relatively (measured* $\sum_k 2D_{C_k}/|F_k| = 1.59 \times 10^{-7}$ *against* $\bar{\chi} \approx 9.16$*). "Distance to the component boundary" is thus* $\geq 1 - 8.87 \times 10^{-4}$ *in w-units — the*

quantitative form of "never jumps components". Scope: structural part is general; the measured figures are 1D. The sequence $\{C_k\}$ *is the §6 majorant-field correction sequence viewed at step level: each fitted gauge is literally a majorant correction,* $M_{G+C_k,H+C_k}(z|x_k) = M_{G,H}(z|x_k) + D_{C_k}(z, x_k)$ *pointwise (verified to* 5.7×10^{-15} *on the 1D run), splitting as* $C_k = \langle e_k, \cdot\rangle + \frac{\sigma_k^*}{2}(\cdot - x_k)^2$ *— the tilt* e_k *is the model-to-iterate gap of Theorem 5.12, the proximal part* σ_k^* *corrects solver inexactness and restores backward-exactness to one rounding quantum (max* σ*-match* 5.96×10^{-8}*). One object, two languages: field (§6) and step (§8) [Proved — agent level; pending author review; Lean verification pending.].*

(f) ***[OBSERVED, range]*** *The fit is impossible when* $u\|G\| \gg dM$*: DCA is* conditionally backward-stable*. The* stability criterion $u\|G\| \lesssim dM$ *sets the* backward-validity range*, not* χu*;* χu *governs the* forward *evaluation of* $F = G - H$ *(thesis Lemma 5.5.3, where the pair is written* (P, Q)*), whose* rounding-noise scale *is* $u\|G\|$*. Unlike linear-system solves, DCA's backward-validity range is set by the subproblem's signal-to-noise.*

Remark 8.2 (Corrections: what is not proved)**.** (i) *Unconditional exactness is false.* The exactness in (b) is conditional on the fit existing. It need not: for $G(z) = z^2/2$, $H = 0$, $x = 0$, every convex differentiable common gauge C gives the gauged subproblem $\frac{1}{2}z^2 + D_C(z, 0)$ with unique minimizer $z = 0$; any computed $\delta \neq 0$, however small, is *not* backward-exact for *any* common convex gauge. This is an algebraic obstruction, not a grid artifact. (ii) *A realized walk is not proved.* Local fits C_k are relative to the baseline pair, but the DCA step at the cumulative object $(G + \Sigma_k,\ H + \Sigma_k)$ — with $\Sigma_k = \sum_{j<k} C_j$ the formal cumulative gauge of (e), so the pair for the step $x_k \to x_{k+1}$ accumulates the fits strictly before k — sees every $D_{C_j}(\cdot, x_k)$, not just D_{C_k}. E.g. $G = z^2/2$, $H = 0$, $x_0 = 1$: local fits $C_0 = z^2/2$ ($x_1 = 1/2$) and $C_1 = 0$ ($x_2 = 0$) explain the computed run against the baseline, but DCA from x_1 at the cumulative pair $(z^2, z^2/2)$ gives $x_2 = 1/4 \neq 0$. Refitting at each step against the cumulative pair is the *refit operator* $\mathcal{T}\colon \Sigma \mapsto \Sigma + \tilde{C}(\Sigma)$, where $\tilde{C}(\Sigma)$ is the total fit-or-zero gauge along the DCA run on the pair $(G + \Sigma, H + \Sigma)$ (this is the "cumulative-pair refit" object, now named). A fixed point of $\mathcal{T}$ is exactly a gauge at which every step is realized: in all n, a progressing run is realized ($\delta_{\text{struct}} = 0$, the structural yardstick) iff all $Q_k = 0$ iff the fixed-point condition holds along the trajectory. The question closes in the negative: no non-degenerate fixed point exists along trajectories from 0 — the refit iteration diverges (positive feedback through damping; see the method caveat in Theorem 8.1(d)) — and the only fixed point found is a frozen-dynamics artifact (a single mass $M \gtrsim 3 \times 10^5$ freezing the float32 run with every $\tilde{\sigma} = 0$, vacuous). [Proved — agent level; pending author review; Lean verification pending.] The realized-walk question remains open with this negative answer. (iii) The cumulative Bregman work R_K is kept as

the sequence's composable cost; calling it a Lawvere length would need the state-enriched category marked in §5.

Remark 8.3 (Cumulative pairs are gauge arrows, not new objects)**.** The cumulative pair $(G + \Sigma_k, H + \Sigma_k)$, with $\Sigma_k = \sum_{j<k} C_j$ the formal cumulative gauge of (e), is the base pair transported by a gauge arrow of §5 (a sum of gauge arrows is a gauge arrow): its DCA majorant satisfies $M_{G+\Sigma_k,H+\Sigma_k}(z|x_k) \equiv M(z|x_k) + D_{\Sigma_k}(z, x_k)$ (exact algebra; the tilts cancel between the function value and the linear term), verified to 1.4×10^{-13} on the 2D run [Proved — agent level; pending author review; Lean verification pending.]. Hence refitting a gauge $\tilde{C}_k$ on the cumulative pair for backward-exactness is appending one more gauge arrow $(G+\Sigma_k, H+\Sigma_k) \to (G + \Sigma_k + \tilde{C}_k, H + \Sigma_k + \tilde{C}_k)$ with majorant correction $D_{\tilde{C}_k}(\cdot, x_k)$. At the exact level the refit degenerates ($\tilde{C}_k \equiv 0$, by the first-order condition at the exact cumulative minimizer); at the computable level the refit has described content (80-step 2D run: 14/80 feasible versus 19/80 baseline; worst $\|\tilde{Q}\|_{\mathrm{op}} = 61.3$ at $\tilde{D} = 3.1 \times 10^{-14}$; computable DCA steps on the cumulative pair within 2.15×10^{-7} of the baseline, i.e. at the rounding quantum) [Proved — agent level (numerical); pending author review; Lean verification pending.]. Whether the self-referential sequence $\Sigma_{k+1} = \Sigma_k + \tilde{C}_k$ (iterating at object level in the §5 category) closes the realized gap stays [Open], now stated in exact object language.

Experiments. On the 1D double well $F = x^4 - 2x^2$ with Newton subproblems in float32, every step is backward-exact to rounding (median σ-match error 4.8×10^{-11}, max 9.7×10^{-8} vs the float32 quantum 6×10^{-8}); on large steps $\sigma^* = 0$, i.e. no gauge was needed at all; the float32 trajectory tracks float64 to 1.5×10^{-7}. On the ReLU chain (thesis (5.18)) with a derivative-free solver, three regimes appear: (A) while the solver progresses, the slogan holds at solver tolerance; (B) at kink stalls the fitted gauge is large ($\sigma^* \sim 10^5$–10^7) and the quantitative form weakens; (C) when $\chi u_{32} \gg 1$ ($L = 6$) the run diverges and no gauge explains the steps — the subproblem contains only rounding noise. The tilt $e = \tilde{s} - s$ is $O(u)$ throughout, with a small systematic bias (honest caveat on the "random walk" phrasing).

Categorical payoff. The per-step fitted gauges form a gauge-fit sequence (gauge arrows add, §5): a float32 DCA run determines the formal cumulative gauge Σ_K. Small gauges cannot leave the weak component, so backward error never fixes conditioning — Wilkinson's observation, now precise for DCA: the doubly-exponential χ of the naive component is invisible to the backward view, which is exactly why *deliberate* gauge-jumping (adaptive-gauge DCA, §6) is a stronger operation than rounding. This also explains the numerics of §6: naive DCA converges in 1–2 steps despite $\chi \sim 10^{16}$ (steps are backward-exact) but has no control over the forward error (the F-evaluation carries the forward error χu).

Novelty. Two independent literature searches (backward error × DCA/difference-of-convex; DC programming × numerical stability/rounding) returned zero relevant hits: backward-error analysis of DCA appears to be new. Nearest existing instances of the "exact on perturbed problem" pattern: Beuzeville–Buttari–Gratton–Mary (computed neural-network output = exact output of perturbed weights — the forward pass, not an iterative optimizer) and inexact-DCA theory (forward error bounds on inner subproblems — a different mechanism).

Remark 8.4 (Honest ledger for §8)**.** (a),(b) proved (conditional); (c),(f) observed; (d) proved 1D; (e) structural + measured 1D. Unconditional exactness is false; any Lawvere-length reading is still marked open. The realized-walk question is closed in exact form: a progressing 1D run is realized — each step a DCA step of the cumulative pair $(G+\Sigma_k, H+\Sigma_k)$ with $\Sigma_k = \sum_{j<k} C_j$ — if and only if every fitted gauge is affine ($\sigma_k^* = 0$ for all k) [Proved — agent level; pending author review; Lean verification pending.] In n dimensions, all-affine ($Q_k \equiv 0$) is sufficient, and $(\bar{Q}_{<k} - Q_k)\hat{d}_k = 0$ is necessary, where $\bar{Q}_{<k} := \sum_{j<k} Q_j$ and $\hat{d}_k := d_k/\|d_k\|$ (for $d_k \neq 0$). The naive claim that baseline-fitted gauges always realize is false: the exact counterexample $G = z^2/2$, $H = 0$, $x_0 = 1$ with fits $C_0 = z^2/2$, $C_1 = 0$ deviates by $\delta_0 = 1/2$ and $\delta_1 = 1/4$ (verified exactly) [Refuted], where δ_k is the deviation of the computed x_{k+1} from the exact DCA step of the cumulative pair $(G + \Sigma_k, H + \Sigma_k)$ at x_k. On the real 1D and 2D double-well runs the walk is realized to rounding noise ($\delta_{\text{real}} \lesssim 10^{-7}$, the float32 quantum), because every fitted gauge has work $D_{C_k} \lesssim u$. The ledger closes in the negative in all n: realized (structural yardstick $\delta_{\text{struct}} = 0$) $\iff$ all $Q_k = 0$ $\iff$ the refit fixed-point condition holds along the trajectory [Proved — agent level; pending author review; Lean verification pending.]; no non-degenerate self-consistent gauge exists along trajectories from 0 (the refit iteration diverges); the only fixed point is the frozen-dynamics artifact. General n beyond this characterization: open. Refit contraction status: at exact level $\mathcal{T} = \mathrm{id}$, so $\mathcal{T}$ is not a strict contraction in any metric (a strict contraction has at most one fixed point, but here every gauge is fixed — a three-line argument). At computed level $\mathcal{T}$ is not nonexpansive (witness: the mass seminorm expands by $\times 2.06$), so the Banach fixed-point route is unavailable. [Proved — agent level; pending author review; Lean verification pending.] *Feeds hub:* the conditional backward-error certificate is the quantitative leg of the hub's DCA–sheaf thread; the full development is the companion paper [24], whose feasibility certificate is the per-step existence check used here.

9. Kink divisors: a tropical vocabulary for gauge arrows

The purpose of this section is to translate gauge arrows into kink divisors, to prove the PL case, and to mark exactly where the analogy stops. The language comes from tropical geometry (Mikhalkin–Zharkov [19], cited for

language only); the new observation is that convexity mod affine is effectiveness of the kink divisor. We warn the reader that the kink divisor is local while convexity is global — effectiveness is necessary but not sufficient in general.

Idea. Work on $\mathbb{R}$ throughout this section (the claims need kinks in the interior; on bounded or disconnected domains the "affine" statements below must be read locally). Adding a convex gauge C to both halves of $F = G - H$ is, tropically, adding an *effective principal divisor*. Here $\operatorname{div}(f) = \sum_p (f'(p^+) - f'(p^-)) \cdot (p)$ for f piecewise-smooth on $\mathbb{R}$ with finitely many kinks — the *kink divisor*, the weighted list of kink jumps. "Principal" means it comes from a function; "effective" is the tropical geometers' word for nonnegative coefficients.

Then div is a group homomorphism (kink jumps add at each p). Affine functions are exactly the tropical units ($\ker \operatorname{div}$ = affine, PL case — a kink-free PL function is affine; this reframes thesis Thm 5.6.4(ii) (Theorem A.2) — the affine part of the gauge is free). PL convex C mod affine *bijects* with effective principal divisors (proved: injectivity by kink-freeness, surjectivity via $\sum (n_p/2)|\cdot - p|$; 20/20 random PL trials). A gauge arrow $P_1 \to P_2$ exists iff their difference $D = G_2 - G_1 = H_2 - H_1$ is convex mod affine iff (PL case) $\operatorname{div}(D)$ is effective (see *Obstruction* below).

The effectiveness relation is a preorder, not an equivalence relation: it is reflexive and transitive but *not* symmetric. The witness is explicit: for the pair (x^2, x^2) vs $(|x|, |x|)$ on $F = 0$, the difference $D = |x| - x^2$ has $\operatorname{div}(D) = +2 \cdot (0)$ while $\operatorname{div}(-D) = -2 \cdot (0)$ — the kink jump of $|x|$ at 0 is $1 - (-1) = 2$, the model case $\operatorname{div}(|x|) = 2 \cdot (0)$. The $+2 \cdot (0)$ coefficient is effective and the $-2 \cdot (0)$ one is not, so the preorder records one direction as "adding" and not the other. This witnesses *asymmetry of the effectiveness preorder*, not a gauge arrow: $D = |x| - x^2$ is not convex mod affine, so no gauge arrow exists in either direction (see *Obstruction*).

The objects reachable from P form its principal up-set $\uparrow P$, consistent with Theorem 5.2 (arrows only add convexity). The effectiveness preorder strictly refines tropical linear equivalence, in one direction only (this is the "asymmetric sense"): it treats the two directions of a pair differently. Its group completion — the equivalence relation the preorder generates — recovers tropical linear equivalence. In the PL case every f is DC, so every $\operatorname{div}(f) = \operatorname{div}(C_1) - \operatorname{div}(C_2)$ is a zigzag, a length-2 path through a common bound, not a symmetrization: $x \sim y \iff x \le y \wedge y \le x$ would force $D_1 = D_2$ mod affine, which is strictly stronger.

Obstruction. If $\operatorname{div}(D)$ is not effective, no gauge arrow $P_1 \to P_2$ exists (proved: a gauge arrow gives $D = C + \ell$ with C convex, so every kink jump of D is ≥ 0); for PL D this is also sufficient.

On $F = 0$: $(x^2, x^2) \not\leftrightarrows (|x|, |x|)$; one direction is obstructed by $-2 \cdot (0)$ (a negative kink — not effective), the other by smooth concavity ($D = |x| - x^2$ has $D'' \equiv -2$ on each ray; the Jensen gap $D((x+y)/2) - (D(x) + D(y))/2$ is 90.0 at $x = -10$, $y = 10$). *Weak components vs arrows (scope).* Absence

of arrows is not absence of weak connection (§5): with *full* convex gauges, (x^2, x^2) and $(|x|, |x|)$ are weakly connected through $P_* = (x^2 + |x|,\ x^2 + |x|)$ (gauges $|x|$ and x^2); with *smooth-only* gauges the zigzag is obstructed and the $|x|$ kink is the separating singularity invariant. Every component claim in this section carries its gauge-class scope: PL, smooth, or full-convex.

The honest summary: the kink divisor is *local* — it registers only kinks, not smooth (non-kink) behavior — while convexity is *global*; effectiveness is necessary but not sufficient in general. The signed kink divisor $\delta(G, H) = \mathrm{div}(G) - \mathrm{div}(H)$ is gauge-invariant but not complete ($\delta = 0$ for both the P_1/P_2 pair and the $(x^2, x^2)/(|x|, |x|)$ pair), so no divisor-class invariant separates the reachable sets: the principal up-sets are genuinely finer than divisor classes, in one direction only.

Honest ledger. The disconnectedness of gauge components is *analogous* to the fibered structure of the tropical Picard fibration $\bigsqcup_d \mathrm{Pic}_d$ — this is analogy only, not a theorem about gauges: the gauge arrows form a monoid, not a group; there is no integer degree, no compact curve. The Mikhalkin–Zharkov theorems (Abel–Jacobi, Riemann–Roch, theta) do *not* transfer; we cite [19] for the language, never as proving anything about gauges. The gauge–sheaf endofunctor of §7 tropically (where $\star_t = +/-$) is tensoring both factors with the *effective* tropical line bundle $\mathcal{E}_C$ — the qualification is essential (the claim fails without it): a general line-bundle twist would allow non-effective C. Open: a compact tropical completion making the global obstruction finite, and any Riemann–Roch-type statement for the gauge category (not even clear what h^0 would count).

Remark 9.1 (Honest ledger for §9)**.** PL kink calculus proved; effectiveness necessary but not sufficient in general (kink divisor is local, convexity global). The $(x^2, x^2)/(|x|, |x|)$ witness is for preorder asymmetry, not an arrow; weak-component claims carry gauge-class scope. Picard language is analogy only.

10. Face-words on the cube: a numerical experiment

The purpose of this section is to test a macroscopic prediction, to exhibit the exact failure mode of the quantitative bound, and to record the experiment's limits. The gauge-fit sequence of §8 is a per-step, microscopic phenomenon. At the macroscopic scale, the natural discrete observable of a penalized gradient flow is its *face-word*: the compressed sequence of face codes visited by the projected trajectory $P_K(x_\varepsilon(t))$ — the paper's coarse record of which faces the trajectory touches. We report a numerical experiment testing whether face-words oscillate on the cube $K = [0, 1]^3$: 1200 trajectories of the penalized flow

$$\dot{x}_\varepsilon \;=\; -\nabla\varphi - \varepsilon^{-1}(x - P_K(x)), \tag{19}$$

(The penalized energy is $\Phi_\varepsilon(x) = \varphi(x) + \frac{1}{2\varepsilon}\|x - P_K(x)\|^2$, i.e. the flow is $\dot{x}_\varepsilon = -\nabla\Phi_\varepsilon$.) with φ a quadratic saddle chosen by automated search

to visit all strata: 12 KKT points across 4 strata (interior saddle, 4 face saddles, 2 edge minima, 3 edge maxima, 2 vertex minima; global minimum $\varphi = -1.301$ at $(1, 0, 1)$). 300 initial conditions (55% interior, 15% on faces, 10% near the interior saddle, 10% near a face saddle, 10% outside K) times $\varepsilon \in \{0.05, 0.02, 0.01, 0.005\}$.

No oscillation, ε-independent statistics [reported numerical evidence — not independently reproduced]. The mean number of face-transitions is 2.89–2.92 across the four ε; the maxima are 4, 5, 5, 6 for $\varepsilon = 0.05$, 0.02, 0.01, 0.005 respectively. No face is ever visited 3+ times; 45/1200 trajectories show a single touch-and-return, never a repeated 2-cycle; all 1200 terminate at the two vertex minima. Face-words stabilize across ε (48–50/50 identical on fixed initial conditions). Qualitative no-oscillation holds with a wide margin, with no cube-specific combinatorics needed.

[Evidence status: reported numerical evidence. (a) The headline numbers above (means 2.89–2.92, maxima 4/5/5/6, the 45/1200 touch-and-returns, the 48–50/50 ε-stability, and the 271/300 figure in the *Limitation* paragraph below): the code, the actual quadratic matrix, initialization samples, tolerances and raw data are not in the packet; these figures were reported by the agents who ran the experiment and are not independently reproduced — they cannot be re-run from this packet. (b) An independent protocol — our own saddle matrix, initial conditions, and tolerances, in scripts that *are* in the packet — qualitatively reproduces no-oscillation (no face visited 3+ times; no repeated 2-cycle, at all four ε) and supplies measured per-charged-transition dissipation $c(\delta)$: 0.016/0.024/0.053/0.097 at $\delta = 0.01/0.02/0.05/0.1$ (6–35× the naive 0.0028; 19× at $\delta = 0.05$). The protocol scripts are `agent-notes/assault-a-w5/w5_flow.py` (seeded: `seed=1000+int(eps*100000)`; the script's docstring states explicitly that it is *not* a reproduction of the headline numbers, since the paper's quadratic matrix was never published) and `agent-notes/assault-a-w5/w5_bound.py` (the targeted 300-round random search), with outputs `w5_flow_results.json` and `w5_flow_trans.npy`. These $c(\delta)$ are reported as constants measured per (problem, code), in the spirit of E3, not as universal values; the 300-round targeted search could not push the minimum dissipation below the plateau $0.0514 \approx 0.053$, so the values are stable against sampling, not sampling artifacts [computed — agent level; pending author review; Lean verification pending.].]

Honest caveat: the quantitative bound is vacuous without a macroscopicity threshold [computed]. The energy-counting bound $(B/c)(n+1)+n$ with the naive per-step minimum up-cost $c = 0.0028$ gives 3552–13850 against a reported maximum of 6 (reported, not independently reproduced). The cheapest "up-transitions" are short hops face→interior→adjacent face (verified with the strictness margin $m = 0.44 > 0$ at the jump, yet dissipating only 0.0028); the energy lemma's cost threshold requires the excursion

to *exit* a δ-neighborhood of the face, which a quick hop to an adjacent face never does.

A non-vacuous quantitative bound needs a macroscopicity threshold (no-chatter) plus geometric separation to exclude these hops — the threshold is essential for any quantitative statement. The qualitative prediction is confirmed.

Limitation. This φ's flow is gradient-dominated downhill: all ω-limits are vertex minima, and genuine macroscopic up-excursions are essentially absent (reported, not independently reproduced: 271/300 trajectories have no up-transitions at $\varepsilon = 0.005$). A stronger stress test uses a φ with saddle networks and repeated face visits.

Correction: a *directed* heteroclinic cycle among saddles is impossible for this gradient flow, since $\dot{\Phi}_\varepsilon = -\|\nabla\Phi_\varepsilon\|^2 < 0$ on every nonconstant orbit — energy strictly decreases along connecting orbits. Face-word cycles must not be confused with heteroclinic cycles; the testable targets are non-cyclic saddle networks and repeated face visits.

Remark 10.1 (Honest ledger for §10)**.** Qualitative no-oscillation is reported numerical evidence, not reproduced; the quantitative bound is vacuous without a macroscopicity threshold (failure mode identified: short hops). No directed heteroclinic cycles exist in gradient flow.

11. Open problems

The purpose of this section is to list, with acceptance conditions, what this paper leaves open. Each item is a precise question with a stated status — [OPEN], [RESTRICTED], [SUPERSEDED], or partially closed — not a vague direction.

(1) *Functoriality / internal iteration [PARTIALLY CLOSED].* Parameterized specialization along graph(h) is computed: its fiber is $k_{[0,\infty)}$, hence slope-free as an abstract sheaf (the "i.e." in the §4 gap remark was loose language). The at-point family $\{x \mapsto \nu_{(x,h(x))}(\mathcal{E}_h)\}$ is the right object, and nonzero inter-fiber morphisms exist: for distinct slopes $s_x = \nabla h(x) \neq s_{x'} = \nabla h(x')$ ($H_x = \{(v,\tau) : \tau \geq \langle s_x, v\rangle\}$ the supporting half-space), the canonical composition $k_{H_x} \twoheadrightarrow k_{H_x \cap H_{x'}} \hookrightarrow k_{H_{x'}}$ is nonzero (isomorphisms on stalks at $(0,\tau_0)$, $\tau_0 > 0$) [Proved — agent level; pending author review; Lean verification pending.] What is missing is a sheaf-functorial choice coherent with the iteration. The choice correspondence $x \mapsto \partial h(x)$ is upper hemicontinuous (small moves in x cannot make the subgradient set jump outward) [Proved — agent level; pending author review; Lean verification pending.] but admits no functorial continuous selection (no selection is continuous at a kink). A canonical iteration endofunctor Φ_K with kernel $K = k_{\Gamma_T}$ (Γ_T the graph of the DCA update map T, $\Phi_K(F) = Rp_{2*}(K \otimes p_1^{-1}F)$) iterates the DCA sequence exactly:

$\Phi_K^m(k_{\{x_0\}}) \simeq k_{\{x_m\}}$ at exact level [Proved — agent level; pending author review; Lean verification pending.]; it is canonical (no arbitrary choices) but not purely sheaf-functorial, because the Argmin readout is an external set-theoretic extraction, not a sheaf functor. Still [Open]: a purely sheaf-functorial iteration, and any strict functoriality at computed (float) level — the model-to-iterate barrier blocks it in both directions.

(2) *General n comparison [ONE-STEP CLOSED; iteration OPEN].* The one-step form is closed by Theorem 4.6 after the $\tau > 0$ correction (one-step Theorem A*: subanalytic is now an explicit hypothesis, since specialization needs it; smooth canonical specialization has field-level support). The smooth-case statement of Conjecture 4.12 needs the same subanalytic correction. The internal categorical iteration is not closed.

(3) *Microlocal gauge theorem.* (a) *Microsupport estimates [Proved — agent level; pending author review; Lean verification pending.]* The height-convolution product $\star$ lifts to a sharp microsupport equality for convex data: the named identities are (i) the microsupport-from-tangent-cone lemma, (ii) the slope shift $s' = s + \nabla C$ (Moreau–Rockafellar sum rule), and (iii) sharpness of the estimate; the non-characteristic/properness hypotheses are automatic here (no subanalyticity needed), with a numerical check in the polyhedral case. (b) *Propagation under the iteration kernel* Φ_K (the sheaf-level lift of one DCA step). Three levels: (1) [Proved — agent level; pending author review; Lean verification pending.] for sheaves F satisfying (i) avoidance of the conormal kink (the singular locus where the subdifferential jumps) and (ii) compact support (A-078); (2) [Proved — agent level; pending author review; Lean verification pending.] for skyscraper sheaves (sheaves supported at a single point) at every x_0, including at a kink, with no hypotheses (A-079, via face duality and sharpness); (3) the all-F, no-hypotheses form stays [OPEN] (obstructions A-076 and A-077(b); no counterexample known). *Correction of expectations (A-069).* The non-characteristic hypothesis for the iteration kernel $K = k_{\Gamma_T}$ (the constant sheaf on the graph of the iteration map T) is not genuine: it holds automatically for every F by the graph structure of the kernel (T continuous suffices). The only hypothesis with content is T proper. *Regularity (A-071).* The explicit conormal formula for Λ_K (the Lagrangian correspondence of the iteration kernel, i.e. its microsupport) requires $g, h \in C^2$; at pure C^1 the iteration map T is only continuous. In the paper, either assume C^2 for the microlocal section or keep $\Lambda_K = \mathrm{SS}(K)$ (the microsupport of K) abstract. *Correction of expectations, nonsmooth fan.* Unlike the smooth case (A-069), the non-characteristic condition for the tensor product is genuine here: the breaking covector is explicit, $(\xi, 0)$ on the top stratum $S_{I,I}$ of the kink cell (the standing

fan face $\{x_0\} \times \mathrm{relint}(\partial h(x_0))$, where relint is the relative interior and $\partial h(x_0)$ the subdifferential of h at x_0). Properness genuinely fails too: $p_2|_R$ (the projection to the second factor, restricted to the correspondence R) is never proper for polyhedral h.

(4) *Solver-cost monotonicity [CLOSED — RESTRICTED AS PROVED].* Monotonicity is false in general: with a fixed solver and a fixed cost model, an ill-conditioned convex gauge step can still raise the cost ($\kappa : 100 \to 50100$, actual iteration count $2062 \to 1038226$ on the $\Psi_\lambda = \frac{\lambda}{2} z_2^2$ ray; numerical counterexample). For the certified cost model $\bar{T} = \kappa \log(D/\varepsilon)$ with gradient descent and the step-size rule $\eta_t = 1/L_t$ fixed in the solver spec, monotonicity along a gauge arrow $C_2 = C_1 + \Psi$ at fixed x_k holds iff $\kappa_\Psi \le \kappa_1$: with $\kappa_2 = (L_1 + L_\Psi)/(\mu_1 + \mu_\Psi)$, $\kappa_2 \le \kappa_1 \iff \kappa_\Psi \le \kappa_1$. ($\kappa = L/\mu$ is the subproblem condition number, μ the strong-convexity modulus, L the gradient Lipschitz constant.) In particular a scalar ray of a well-conditioned generator ($\kappa_{C_0} \le \kappa_1$, e.g. $C_0 = \frac{1}{2}\|\cdot\|^2$ with $\kappa_{C_0} = 1$) is monotone nonincreasing. Monotonicity is along the gauge arrow at fixed x_k, not along iterates; "fixed solver" must fix the step-size rule (a literally fixed η can diverge). [Proved — agent level; pending author review; Lean verification pending.]

(5) *Fan-DCA [OPEN].* At a kink, classical DCA commits to one subgradient; specialization exhibits the full fan canonically. The choice correspondence $x \mapsto \partial h(x)$ is upper hemicontinuous [Proved — agent level; pending author review; Lean verification pending.] — small moves in x cannot make the subgradient set jump outward — which supports the multivalued fan form: the kernel formalism is not blocked at a kink. What is blocked is only a functorial single-valued selection (no selection $x \mapsto \sigma(x) \in \partial h(x)$ is continuous at a kink). Design and analyze a branching DCA exploring all rays $s \in \partial h(x_k)$: needs algorithm, selection/exploration rule, descent/convergence, and cost. The $|x|$ example motivates but does not guarantee.

(6) *Microlocal descent [OPEN as microlocal].* Classical majorization descent is known and holds for varying splittings with exact subproblems (Proposition 6.3); a genuinely microlocal proof is the open part.

(7) *Optimal representation.* (a) *Admissible objects and normalization [Proved — agent level; pending author review; Lean verification pending.]* The admissible class $\mathcal{A}_{\bar{\mu},\bar{L}}(F)$: pairs (G, H) of convex functions with $\bar{\mu} I \preceq \nabla^2 G \preceq \bar{L} I$ and $G - H = F$, normalized by the affine-equivalence slices $H(x_0) = 0$, $\nabla H(x_0) = 0$. Here $\bar{\mu} > 0$ and $\bar{L} < \infty$ are uniform lower/upper curvature bounds, and $\sim_{\mathrm{aff}}$ identifies pairs differing by an affine function. (b) *Objective quality [Proved — agent level; pending author review; Lean verification pending.]* Primary: $Q_\kappa = \kappa_G$, the condition number of $\nabla^2 G$ (ratio of largest to smallest

eigenvalue), which orders candidates the same way as the certified subproblem cost. Secondary: $Q_{\mathrm{tr}} = \sup \mathrm{tr}(\nabla^2 G)$ (minimal damping). Excluded: the χ functional, the orbit-dependent R_K as primary, and the thesis's (5.55) (undetermined weights). (c) $F = 0$ *and signs [Proved — agent level; pending author review; Lean verification pending.]* The quality distinguishes (C, C) pairs with $\kappa = 1$ vs. $\kappa = 100$; $(0, 0)$ is excluded by the $\bar{\mu}$ floor; the sign asymmetry is structural (numerical example). (d) *Quadratic case and existence [Proved — agent level; pending author review; Lean verification pending.]* For quadratic F: $\min \kappa_G = 1$, attained exactly by the isotropic pairs $Q = \lambda I$, with the trace tiebreak selecting the unique $\lambda^* = \max(\lambda_{\max}(A), \bar{\mu})$, where $A = \nabla^2 F$ is the constant Hessian. For general F: the upper-smooth characterization holds, and existence is closed by the dichotomy — feasible iff $s \leq \bar{L}$, where $s = \sup_\Omega \lambda_{\max}(\nabla^2 F)$; then $\min \kappa_G = 1$ is attained by an explicit isotropic pair. There is no regime with $1 < \min \kappa_G < \infty$: the only alternative is infeasibility, when F is not $\bar{L}$-upper-smooth. The A-086(ii) compactness blueprint is superseded by this dichotomy (A-088). *Numerical evidence* (`w19_optimal_repr.py`, `ALL OK`): three candidates with $\kappa = 31/2.5/1.0$; the $F = 0$ case gives $\kappa = 1$ vs. 100; signs: $\mathrm{opt}(-F) \neq \mathrm{swap}(\mathrm{opt}(F))$ with the swap at $\kappa = \infty$; the trace tiebreak is strictly increasing; affine-invariance holds. *Lemma (A-087).* The condition number κ (ratio of largest to smallest Hessian eigenvalue) is lower-semicontinuous under weak-$*$ L^∞ convergence of Hessian fields [Proved — agent level; pending author review; Lean verification pending.] Honest scope: this lemma is no longer used for existence (A-088 goes by a direct route); it is kept as an independent lemma. *Correction of the 1D degeneracy note (A-089).* The 1D degeneracy $\kappa \equiv 1$ holds only for quadratic F; a non-quadratic 1D F has genuinely varying κ (e.g. $\kappa = 25$ for a naive pair on $F = x^4 - 2x^2$), while $\min \kappa_G = 1$ still holds whenever feasible. (Numerical; supersedes the unqualified illustration rule that 1D is always degenerate.)

(8) *Canonical gauge normalization [CLOSED except two-sided metric].* v0.2 took gauge arrows on the nose to repair the mod-affine enrichment. The Bregman cost of Theorem 5.6 resolves this cleanly while keeping the thesis's original arrows, so the R2 normalization program is retired. The working object quotient is $\sim_{\mathrm{aff}}$, identifying (G, H) with $(G + \ell, H + \ell)$ for affine ℓ: the affine shifts form a group, so this is a genuine equivalence relation, compatible with gauge arrows (which are already mod affine) [Proved — agent level; pending author review; Lean verification pending.] The naive one-step relation "differ by a common convex gauge" is not an equivalence — it is not symmetric, since the reverse step would need $-C$, concave unless C is affine (e.g. $C(x) = (x - 2.5)^2 - 1$ admits no gauge arrow back). On $\sim_{\mathrm{aff}}$, the descend table is: F, DCA iterates, model residuals, fitted

gauges C_k, cumulative work R_K, the feasibility pattern, the Bregman cost d, the majorant field (a complete invariant of the class), and face-words descend; χ does not [Proved — agent level; pending author review; Lean verification pending.] Face-words of a DCA run are bona fide observables on $\sim_{\text{aff}}$: under $(G, H) \mapsto (G+\ell, H+\ell)$ with ℓ affine, the affine parts cancel in the DCA subproblem, so the iterates — and hence the face codes and the face-word — are unchanged (30/30 runs identical, $\max|\Delta x| = 4.4\times10^{-16}$) [Proved — agent level; pending author review; Lean verification pending.] Caveat: this is a DCA-side statement; for penalized gradient flow an affine shift changes the gradient, so flow face-words do not descend in general. The Lawvere cost d lives on gauge arrows mod affine and induces no metric on face-words — descent is the shared language, not a shared metric [Observation]. χ fails to descend even through a single gauge step: on $F = x^2 - |x|$ over $\Omega = [2, 3]$, the pair $P = (x^2, |x|)$ and its gauge shift by $C(x) = (x - 2.5)^2 - 1$ give $\chi_P(2.5) = 2.333 \to 1.800$, and the affine shift $\ell(x) = 3x + 1$ gives $2.333 \to 6.867$ [Refuted as a quotient function]; corroborated by $\chi : 1.68 \times 10^{12} \to 19$ through incomparability jumps in one weak component. Separation on objects: after $\sim_{\text{aff}}$ the zero-two-sided-distance pathology disappears by definition, and the one-way Lawvere cost d is separated on quotient arrows; a full two-sided metric on objects via zigzag infima is [Open].

(9) *Adaptive-mirror-map convergence [RATES PROVED; ADAPTIVE CHARACTERIZED; INEXACTNESS THRESHOLD PROVED; BENEFIT REFUTED (proximal sense)].* Descent has the one-line proof in the exact scope (Proposition 6.3). Fixed-gauge rates: sufficient decrease, $O(1/K)$ iterate and stationarity bounds hold [Proved — agent level; pending author review; Lean verification pending.] Adaptive gauges C_k: the $O(1/K)$ rate survives under uniform strong convexity $\inf_k \mu_k > 0$ (μ_k the strong-convexity modulus of C_k); stationarity of cluster points needs in addition $\sup_k L_k < \infty$ (L_k the gradient Lipschitz constant of C_k) [Proved — agent level; pending author review; Lean verification pending.] The $\sup_k L_k < \infty$ condition is genuine, not technical: with $L_k = 2^k \to \infty$ (while $\mu_k = 2^k \geq 1$) the run converges to a non-stationary point ($x_k \to x_0/4.768\ldots \neq 0$ for $G(z) = z^2/2$, $H = 0$); the threshold is sharp, $x_k \to 0 \iff \sum_k 1/L_k = \infty$. Inexact subproblems with residual bound $\|r_k\| \leq \varepsilon_k$: $\sum \varepsilon_k^2 < \infty$ preserves the $O(1/K)$ rate fully; roundoff gives the noise floor $O(u\bar{S})$ (u unit roundoff, $\bar{S}$ a uniform roundoff-scale bound along the trajectory) [Proved — agent level; pending author review; Lean verification pending.] Adaptation benefit: proximal damping (adaptive local-curvature gauges) does *not* accelerate DCA — on the 2D double well it costs 3.5× the outer steps and 3.1× the inner Newton work of plain DCA; the gauge is proximal damping, not a preconditioner [Refuted (proximal sense) — agent level; pending author

review; Lean verification pending.]. Novelty vs adaptive regularizers corrected in §6; the incomparability-jump mechanism (splitting choice lowering χ) is separate and remains open. Still [Open]: linear rate for DC, which needs KL exponent 1/2.

(10) *Quantitative perturbation-sequence bounds, general n [PARTIALLY CLOSED].* The companion E3 paper has the local PSD quadratic-fit repair. Uniform solver error bound is available: measured constants, certified, and trajectory-local sharp ($\|x_k^+ - y_k\| \le uS_k/\bar{\mu}_k$ with $\bar{\mu}_k$ the smallest Hessian eigenvalue along the step segment, sharp to 10^{-9}; u unit roundoff, x_k^+ the exact minimizer, y_k the computed step). Cumulative-pair comparison is available: the identity $M_{G+\Sigma_k,H+\Sigma_k} = M + D_{\Sigma_k}$ says a DCA step of the cumulative pair is a gauge step on the base pair (not a new object class), and the fixed-point=realized characterization holds in all n (realized walk $\iff$ all $Q_k = 0$ $\iff$ the refit fixed-point condition). Still missing: a length proof for a defined metric/category (the full two-sided metric on objects is open).

(11) *Non-vacuous face-word bounds [CLOSED at agent level; uniform energy step SKETCH].* The cube experiment of §10 confirms qualitative no-oscillation (as reported evidence) but exhibits the exact failure mode (short hops) that makes the quantitative energy bound vacuous.

Closed up to a structural slack [Proved — agent level; pending author review; Lean verification pending.] The paper's "up-transition" is now defined: a *charged* face transition is one that does not lower the face dimension (up or lateral). (i) No-chatter thresholds: the δ-core of a face — the points at relative distance at least δ from its relative boundary (for gradient flow), recorded in the δ-no-chatter face-word — and a step-length threshold $s_{\min}$ (for DCA: it separates the convergence-flicker chatter, all below 10^{-3} and all on dimension-lowering transitions, from the macroscopic transient; zero word changes for $s_{\min} \in [10^{-9}, 10^{-3}]$ on the tested saddles). (ii) Geometric separation: δ-cores of distinct faces are Euclidean-separated by at least δ (proved, sharp, checked exactly on all 351 face pairs of the cube) [Proved — agent level; pending author review; Lean verification pending.] each charged transition dissipates at least $c(\delta) > 0$ over a macroscopic length $\ge \delta$ — the argument is a [SKETCH] (arc-length parametrization plus compactness; the orbit must enter a ball around an isolated critical point, and linearization shows it still dissipates $\sim r^2 > 0$ there; the uniformization step is still a sketch) — with measured $c = 0.016/0.024/0.053/0.097$ at $\delta = 0.01/0.02/0.05/0.1$, i.e. 6–35× the naive 0.0028 (19× at $\delta = 0.05$). A targeted 300-round random search could not push the minimum dissipation below the plateau $0.0514 \approx 0.053$, so $c(\delta) > 0$ is genuine, not a sampling artifact. (iii) The combinatorial bound

#transitions $\leq (B/c)(n+1)+n$ holds (charged transitions cut the word into strictly dimension-dropping segments of length $\leq n$, with B the energy budget) and is reproduced by the scripts. Valid but still loose (2597 vs 2 observed macroscopic transitions on a global budget B): the remaining slack is structural — the global B versus the rarity of charged transitions — and needs a charged-transition-specific Lyapunov budget [Open]. Do not use directed heteroclinic cycles to stress-test gradient flow.

(12) *Dual DCA in C^*-coordinates [Proved — agent level; pending author review; Lean verification pending.]* With C Legendre (strictly convex C^1 with bijective gradient $\nabla C : \operatorname{int}(\operatorname{dom} C) \to \operatorname{int}(\operatorname{dom} C^*)$) and $g := G + C$, $h := H + C$ Legendre: (1) the correct dual DCA is the Toland shadow (the dual formulation of the DC program) on the absorbing pair (g, h), with dual variable $t_k = s_k + m_k$, where $s_k \in \partial H(x_k)$ is the subgradient and $m_k = \nabla C(x_k)$ the mirror coordinate (A-090); (2) the naive guess via (H^*, G^*) and C^* is refuted — the gauge does not pass through Fenchel duality (F^* denotes the Fenchel conjugate), and coordinate substitution gives a nonconvex subproblem (A-091); (3) dual descent plus primal descent in the y-coordinates (A-092); (4) dual step existence, the mirror of the A-054 condition (A-093). Deliberately still open: nonsmooth H (the fan case), the dual backward error, and transfer of the sheaf-level results to the dual side. *Honest scope.* The dual DCA is a theoretical shadow, not an independent algorithm: the step $t_k \to t_{k+1}$ needs x_{k+1} from the primal. Everything "for free" is at theorem level. In the paper, speak of "the dual sequence $\{t_k\}$", never of "running the dual DCA". The original questions are now answered: "C Legendre and the dual domain" — $\nabla C : \operatorname{int}(\operatorname{dom} C) \to \operatorname{int}(\operatorname{dom} C^*)$ is bijective and $m_k \in \operatorname{int}(\operatorname{dom} C^*)$ (numerical: $(0,\infty)^2$ for the exponential gauge, $(-\infty, 0)$ for the log-barrier — both proper subsets of $\mathbb{R}^n$); "comparison theorem and step existence conditions" — A-092 (comparison/descent) and A-093 (step existence).

(13) *Lagrangian/sheaf counterpart [OPEN].* The admissible categories are now stated explicitly: on the sheaf side, $D^b_{\mathbb{R}\text{-c}}(X \times \mathbb{R}_t; \tau \geq 0)$ (the bounded derived category of $\mathbb{R}$-constructible sheaves on $X \times \mathbb{R}_t$ with microsupport in $\tau \geq 0$, where τ is the extra height coordinate), with objects the epigraph sheaves $\mathcal{E}_f = k_{\operatorname{epi}(f)}$ (k_Z the constant sheaf on Z), specializations, and kernels k_R; on the Lagrangian side, exact Lagrangians in T^*X generated by the graphs Γ_{-df} plus conic conormal fans (the polyhedral case), with the wrapped setting noted where f is not compactly supported (A-095). The smooth dictionary and the iteration correspondence are closed at agent level (A-095/A-096) [Proved — agent level; pending author review; Lean verification pending.] Warning (A-096, naive guess refuted): specialization gives the Lagrangian of the subderivative (the first-order model), not the

tangent cone of the Lagrangian. Nadler–Zaslow does not by itself supply the iteration (five named obstructions, A-097). The iteration and fan parts reduce to hub Conjecture 2 (singular Floer=Morse for the DCA correspondence) (A-098); that hard core is open, so the [OPEN] label is kept. *Warnings for the writer.* Cite A-096 as the "naive guess refuted" warning: specialization is the Lagrangian of the subderivative, not the tangent cone of the Lagrangian. Cite A-098 for "Nadler–Zaslow does not by itself supply the iteration": five named obstructions (A-097) — (O1) admissibility (the fiberwise sum only closes on graphs), (O2) choice externality (no functorial branch selection), (O3) readout min-vs-critical ($\tau > 0$ selects minima; Maslov index 0 only in the Morse case), (O4) iteration endomorphism (Nadler–Zaslow is an equivalence of categories, not of dynamical systems), (O5) level mismatch (the paper lives at the conified Tamarkin level; the honest Lagrangian counterpart is at the J^1 contact level). *Fukaya honesty note.* The Fukaya side adds no new theorems to paper A; its value is (i) an explicit, checkable dictionary, (ii) locating §11(13) inside hub Conjecture 2, and (iii) a concrete, falsifiable test case for singular Floer=Morse (the DCA correspondence Λ_T on fiber branes vs. DCA/Fan-DCA on points).

What this paper contributes. In one page: the DCA step is lifted to three sheaf moves — specialization, height convolution, argmin readout — and the one-step comparison is proved in all dimensions with the corrected $\tau > 0$ readout (Theorems 1.1 and 4.6); the standing gap is the internal iteration, which needs functoriality of specialization along graph(h). Decomposition-dependence is categorified as a gauge groupoid and preorder, enriched by the affine-invariant Bregman cost, with quality functionals as functors on the $C \geq 0$ subcategory (§5); the gauge is identified as the Bregman generator of a composite mirror-descent step (§6) and as a sheaf convolution whose readout is the Bregman divergence (§7). The computed run is backward-exact on a gauge-perturbed pair, with the gauge-fit sequence and its cumulative Bregman work measuring the rounding cost (§8); the tropical reading and the face-word experiment (§§9–10) test the picture at the kink and at macroscopic scale. Everything not proved is marked open in place with acceptance conditions. The hub's DCA–sheaf thread takes the proved one-step half from here; the missing half is gap G5.

Appendix A. Cited results

This appendix quotes, in full, the results from the companion works that this paper cites by theorem number. Theorem numbers shift across versions; the statements below are the content cited.

Theorem A.1 (Common convex gauge adds a Bregman term; Proposition 5.5.1 of the monograph [12])**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open and convex, let*

$F = G - H$ on Ω with G, H convex and H differentiable, and let $C : \Omega \to \mathbb{R}$ be convex and differentiable. Write the classical DCA majorant at $x \in \Omega$ as

$$M_{G,H}(z \mid x) := G(z) - H(x) - \langle \nabla H(x),\, z - x\rangle, \qquad z \in \Omega.$$

Then for the equivalent decomposition $F = (G + C) - (H + C)$,

$$M_{G+C,\,H+C}(z \mid x) \;=\; M_{G,H}(z \mid x) \;+\; D_C(z, x), \qquad z, x \in \Omega, \tag{20}$$

where $D_C(z, x) = C(z) - C(x) - \langle \nabla C(x),\, z - x\rangle$ is the Bregman divergence of C.

Used in §§5–8: a gauge arrow C sends the DCA majorant to $M_{G,H} + D_C$, so gauge change is a proximal penalty on the subproblem.

Theorem A.2 (The bit-optimal gauge inside a strong-convexity class; Theorem 5.6.4 of the monograph [12])**.** *Let $\Omega \subseteq \mathbb{R}^d$ be nonempty, open, bounded and convex, and let (G, H) be an NDC pair on Ω for $F = G - H$ in the sense of Definition 5.2.1 of the monograph, with H differentiable. Assume*

$$f_{\min} := \inf_{\Omega} |F| > 0, \qquad F_{\max} := \sup_{\Omega} |F| < \infty, \qquad \bar{\chi}_0 := \sup_{\Omega} \frac{G + H}{|F|} < \infty. \tag{21}$$

Let $\Phi : \Omega \to [0, \infty)$ be differentiable, convex and μ_Φ-strongly convex with $\mu_\Phi > 0$, let $\rho > 0$, and for $\sigma \in (0, \rho\mu_\Phi]$ define the admissible gauge class

$$\begin{aligned} \mathcal{A}^{\mathrm{g}}_\sigma := \big\{ C : \Omega \to \mathbb{R} \;:\; & C \text{ differentiable, convex, } C \geq 0, \\ & C \text{ is } \sigma\text{-strongly convex, } \rho\Phi - C \text{ convex} \big\}. \end{aligned} \tag{22}$$

For $C \in \mathcal{A}^{\mathrm{g}}_\sigma$ write χ_C for the subtraction condition number (5.16) of the monograph for the shifted pair $(G + C, H + C)$, $\chi_0 := \chi_{(G,H)}$, and

$$\Xi(C) := \sup_{z \in \Omega} \big(\chi_C(z) - \chi_0(z)\big) \qquad \textit{(the } \text{excess conditioning} \textit{ of the gauge).} \tag{23}$$

Then:

(i) Exact bit identity. *$(G + C, H + C)$ is an NDC pair for F, $G + H > 0$ on Ω, and for every $z \in \Omega$*

$$\chi_C(z) = \chi_0(z) + \frac{2C(z)}{|F(z)|} = \chi_0(z)\Big(1 + \frac{2C(z)}{G(z) + H(z)}\Big), \qquad \Xi(C) = 2 \sup_{z \in \Omega} \frac{C(z)}{|F(z)|}. \tag{24}$$

The gauge therefore costs exactly $\log_2\big(1 + 2C(z)/(G(z) + H(z))\big)$ extra mantissa bits at z in the guarantee of Lemma 5.5.3 of the monograph.

(ii) The affine part of the gauge is free. *$M_{G+C,H+C}(\cdot \mid x)$ depends on C only through D_C, hence only on C modulo affine functions. For every $x \in \Omega$ the re-centred gauge $C^x := D_C(\cdot, x)$ lies in $\mathcal{A}^{\mathrm{g}}_\sigma$, generates the identical DCA model $M_{G+C^x,H+C^x}(\cdot \mid x) = M_{G+C,H+C}(\cdot \mid x)$, and satisfies $\chi_{C^x}(x) = \chi_0(x)$. Thus the additional bit cost at any one prescribed point can be removed entirely at no cost in the model.*

(iii) The uniform problem has a unique centre and a smallest solution. *Put*

$$\psi(c) := \sup_{z\in\Omega} \frac{\|z-c\|^2}{|F(z)|} \quad (c\in\mathbb{R}^d), \qquad W := \inf_{c\in\mathbb{R}^d} \psi(c). \tag{25}$$

Then ψ is finite, $(2/F_{\max})$-strongly convex and coercive, so it attains its infimum at a unique point c^*, the $|F|$-weighted Chebyshev centre of Ω, and $W = \psi(c^*) \in (0,\infty)$. Moreover

$$\min_{C\in\mathcal{A}^{\mathrm{g}}_\sigma} \Xi(C) \;=\; \sigma W, \tag{26}$$

the minimum being attained by $C^*_\sigma(z) := \frac{\sigma}{2}\|z-c^*\|^2$. Every minimiser C satisfies $C \geq C^*_\sigma$ pointwise on Ω, so C^*_σ is the smallest optimal gauge.

(iv) Exact conversion factor; no interior optimum. *Let $m_0 \geq 0$ be a constant for which G is m_0-strongly convex, the value $m_0 = 0$ (mere convexity) being allowed. The DCA model $M_{G+C,H+C}(\cdot \mid x)$ is $(m_0+\sigma)$-strongly convex for every $C \in \mathcal{A}^{\mathrm{g}}_\sigma$, and the set of pairs (model strength, excess conditioning) achievable as σ ranges over $(0,\rho\mu_\Phi]$ and C over $\mathcal{A}^{\mathrm{g}}_\sigma$ is $\{(m_0+\sigma,\Xi) : \sigma\in(0,\rho\mu_\Phi],\ \sigma W \leq \Xi < +\infty\}$, together with the pairs $(m_0+\sigma,+\infty)$, $\sigma\in(0,\rho\mu_\Phi]$, if and only if $\sup_\Omega \Phi = +\infty$, the value $\Xi = +\infty$ occurring exactly for gauges unbounded on Ω. Its Pareto frontier is the straight segment $\Xi = (m-m_0)W$, $m \in (m_0, m_0+\rho\mu_\Phi]$: the conversion factor is exactly W units of excess conditioning per unit of model strength, uniformly, so no interior σ is distinguished and every linear scalarisation is optimised at an endpoint of the closed segment. In bits, the optimal gauge costs at most*

$$\log_2\Big(1+\frac{\sigma W}{\bar\chi_0}\Big) \tag{27}$$

more than the ungauged pair, so the strength purchased by exactly k extra bits is $\sigma_k = (2^k-1)\bar\chi_0/W$.

(v) Efficiency of the extremal gauge. *Assume in addition $\sup_\Omega \Phi < \infty$. The maximal gauge $C = \rho\Phi$ of Proposition 5.6.3 of the monograph lies in $\mathcal{A}^{\mathrm{g}}_{\rho\mu_\Phi}$, and its* gauge efficiency *$\eta(\rho\Phi) := \rho\mu_\Phi W/\Xi(\rho\Phi)$ lies in $(0,1]$. Moreover $\eta(\rho\Phi) = 1$ if and only if $\rho\Phi$ is itself a minimiser of (26) at $\sigma = \rho\mu_\Phi$, that is, if and only if $\sup_\Omega \Phi/|F| = \mu_\Phi W/2$.*

Used in §5 to bound the within-up-set optimum (part (iv): the Pareto frontier is a straight line).

Theorem A.3 (The alignment estimate in n dimensions, parts (a)–(b); Theorem 4.1(a)–(b) of [24])**.** *Convention. $r := s - \nabla G(y)$ is the corrected residual of §3 of the companion paper and the fit equation is $Qd = r$ throughout. Fix a step with $d := y - x \neq 0$, θ the angle between d and r, $c := \cos\theta = d^T r/(\|d\|\|r\|)$, and $\rho \geq \|r\|$ the roundoff noise scale.*

(a) Fit coefficient. *The constrained least-squares fit* $\sigma_{\mathrm{LS}} := \arg\min_{\sigma\geq 0}\|\sigma d - r\|$ *is* $\sigma_{\mathrm{LS}} = \max(d^T r/\|d\|^2, 0)$, *with misfit* $\|r-\sigma_{\mathrm{LS}} d\|^2 = \|r\|^2 - \max(d^T r, 0)^2/\|d\|^2$ *— in the acute case* $d^T r \geq 0$ *this is* $\|r - \sigma_{\mathrm{LS}} d\| = \|r\|\sin\theta$*; exact fit* $\sigma d = r$ *with* $\sigma \geq 0$ *is solvable iff* $r \parallel d$ *and* $d^T r \geq 0$*, and then* $\sigma^* = d^T r/\|d\|^2$ *(the direct* n*-dimensional analogue of* r/d*). In all cases* $|\sigma_{\mathrm{fit}}| \leq \rho/\|d\|$*, with no alignment condition.*

(b) Curvature of the PSD fit *(conditional on* $d^T r > 0$*). For the rank-one certificate* $Q = rr^T/(d^T r) \succeq 0$ *of the PSD feasibility certificate,*

$$\|Q\| = \frac{\|r\|^2}{d^T r} = \frac{\|r\|/\|d\|}{\cos\theta} \lesssim \frac{\rho}{c\,\|d\|}, \qquad c \in (0, 1],$$

and the smallest isotropic gauge dominating Q *(*$\sigma I \succeq Q$*) is* $\sigma_{\mathrm{dom}} = \|Q\|$*, satisfying the same bound.*

Used in §8: the alignment $\cos\theta$ governs only the gauge curvature, not the fit coefficient nor the size bound.

Theorem A.4 (The n-dimensional roundoff estimate; Theorem 4.1 ("On ρ and c") of [24])**.** *The* n*-dimensional roundoff estimate* $\|r_k\| \lesssim uS_k$ *with* $S_k = c_1\|s\| + c_2\|y_k\|\,\|\nabla^2 G(y_k)\|_{\mathrm{op}}$ *holds under the assumption set stated there:* G *is* C^2 *with* $\nabla^2 G \succ 0$ *on the trajectory (smooth steps only); the Newton solver stops by one of four named rules (exact float32 fixed point, relative stagnation, exactly-zero computed gradient, or gradient-stall with stall-faithfulness). Here* u *is the unit roundoff (*$u = 2^{-24}$ *in float32),* $s = \nabla H(x_k)$ *is the exact slope,* y_k *is the computed subproblem minimizer, and* $\|\cdot\|_{\mathrm{op}}$ *is the spectral (operator) norm. The constants* c_1, c_2 *are measured per (problem, code) and certified by the ratio protocol (*$\max_k \|r_k\|/(uS_k) \leq 1$*). At kinks* $\nabla^2 G$ *does not exist and the estimate does not apply; nonsmooth steps need a separate analysis.*

Used in §8: the roundoff estimate $\|r_k\| \leq uS_k$ that feeds the composability bound of Theorem 8.1(d).

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